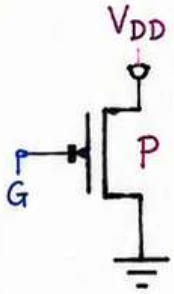
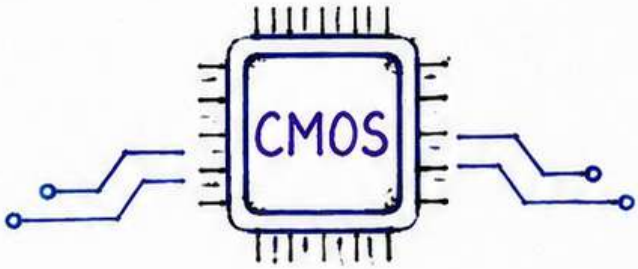
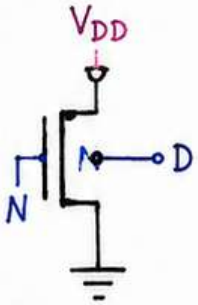
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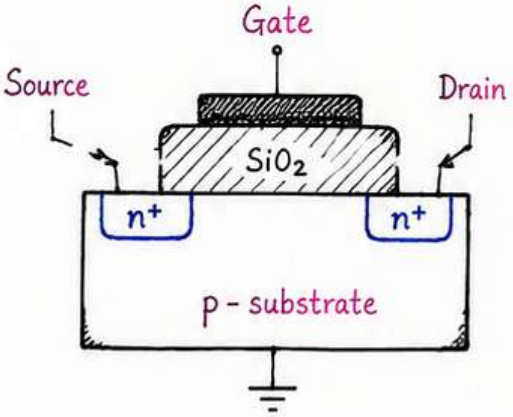


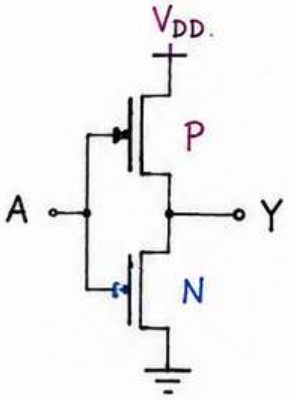


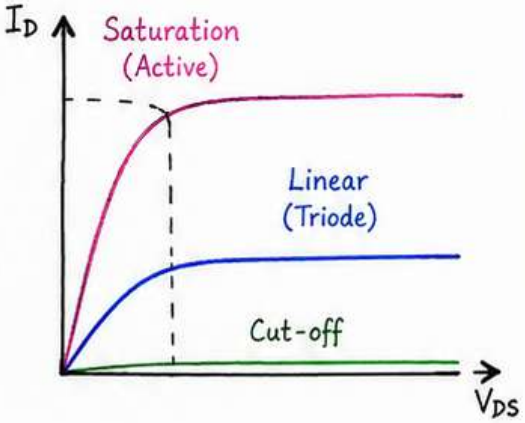
CMOS

– Industry Level Perspective –

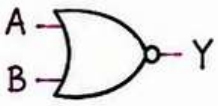








Digital VLSI



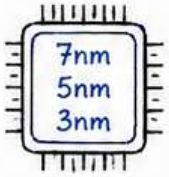
Low Power Design



High Performance Circuits



Scaling to Advanced Nodes

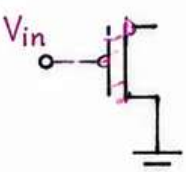


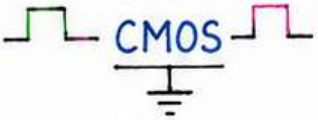
Future Technology

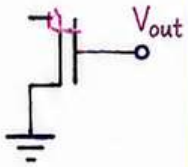


Author :

Konda Tharun Vivekananda

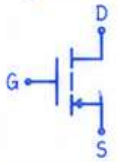






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INDEX



S.No.	Main Heading / Concept	Page No.
1	Introduction to CMOS; Why do we prefer CMOS?; Characteristics of CMOS	1
2	CMOS Technology Overview; Classification of Transistors; MOSFET Applications; BJT vs FET Comparison; Why FET is Preferred Over BJT?	2
3	JFET; MOSFET; JFET vs MOSFET Comparison	3
4	MOSFET Introduction, Terminals, Features & Diagrams; Types of MOSFET – Classification	4
5	N-channel Depletion MOSFET; P-channel Depletion MOSFET – Operation, Diagrams, I-V Characteristics, Transient Analysis	5
6	N-channel Enhancement MOSFET; P-channel Enhancement MOSFET – Operation, Diagrams, I-V Characteristics, Transient Analysis	6
7	Threshold Voltage and its Importance; Applications of CMOS	7
8	CMOS Fabrication using N-well; CMOS Fabrication using Twin Tub	8
9	MOSFET Operating Regions – Cut-off, Linear/Triode, Saturation; Behaviour and Graphs	9
10	MOSFET Channel Effects – Short Channel Effects (SCEs)	10
11	MOSFET Channel Effects – Narrow Channel Effects	11
12	MOSFET Effects – Second Order Effects	12
13	MOSFET Capacitance – Introduction, Diagrams, Oxide Capacitance and Complete Concepts	13
14	MOSFET Capacitance – Lumped Representation; Explanation of Each Capacitance; Effects of MOSFET Capacitances	14
15	CMOS Basic Gates Design – Introduction, Why Design, PUN, PDN, Static Design, Dynamic Design	15
16	CMOS Basic Gates Design – AND Design; OR Design	16
17	CMOS Basic Gates Design – NAND Design; NOR Design	17
18	CMOS Problems and Alternatives – Ratioed Logic, Pass Transistor Logic, Transmission Gate, Differential Pass Transistor Logic	18
19	CMOS – Logic Effort; Calculations, Examples, Advantages/Applications, Logic-Effort Summary	19
20	CMOS Evolution – Why MOSFET Become Difficult; FinFET Introduction	20
21	GAA FET; Comparison of MOSFET, FinFET & GAA FET; Current Industry Technology; Current Semiconductor Trends and Technology	21

CMOS – Industry Perspective

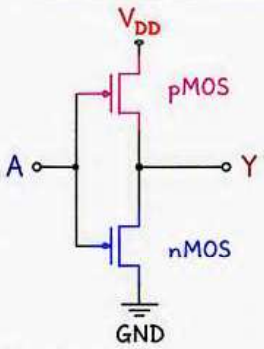
1. Introduction to CMOS

- Definition : CMOS (Complementary Metal Oxide Semiconductor) is a VLSI technology that uses complementary nMOS and pMOS devices.
- Purpose : To implement digital & analog circuits with low power, high noise immunity and high density.
- Fundamental concept : Uses both nMOS and pMOS in pull-down and pull-up networks for efficient switching.
- Importance in VLSI/ECE : Backbone of modern ICs, from microcontrollers to high performance processors and SoCs.
- Real-world applications : Smartphones, CPUs, IoT. devices, Automotive ICs, FPGAs, ASICs, DSPs, image sensors, memories, etc.

2. Why do we prefer CMOS ?

- Very low static power dissipation (ideal case ≈ 0).
- High noise immunity and wide logic swing.
- High packing density → small area.
- Excellent scalability to deep sub-micron nodes.
- Low cost due to simple fabrication (standard CMOS flow).
- Works over a wide supply voltage range.
- High reliability and long battery life for portable systems.
- Compatible with mixed-signal and RF integration.
- Huge ecosystem, EDA tools and foundry support.

CMOS Inverter Structure

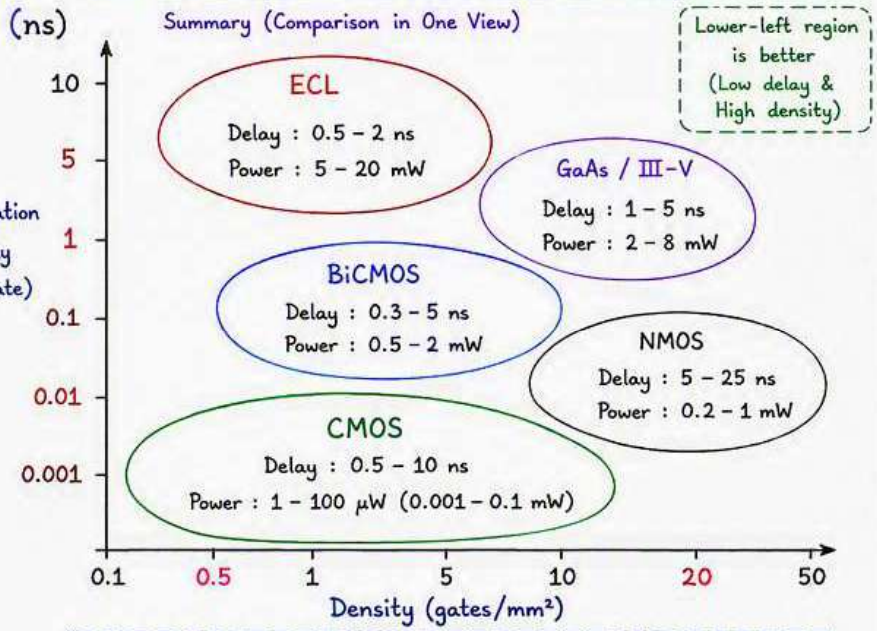


A	Y
0	1
1	0

Structure :

- pMOS connected to V_{DD} (pull-up path)
- nMOS connected to GND (pull-down path)
- Input A drives both gates.
- Output Y is taken at the junction.

Industry Perspective Graph : Delay vs Density

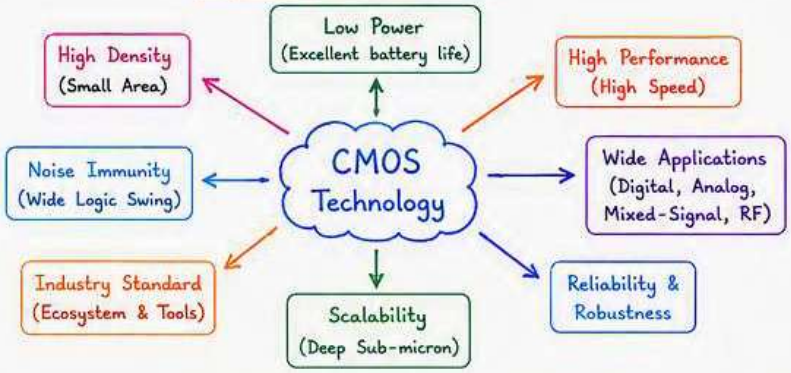


CMOS provides the best delay-density tradeoff for large-scale VLSI integration with very low power. ☆

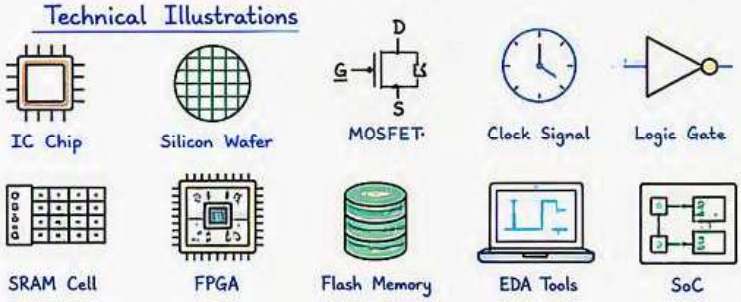
2. Characteristics of CMOS

- Low Power : Almost zero static power, low dynamic power.
- High Noise Immunity : Full rail-to-rail outputs.
- High Density : Complementary structure → small area.
- High Speed (Moderate to High) : Suitable for high performance digital circuits.
- Scalability : Well suited for deep submicron and FinFET nodes.
- Reliability : Good reliability and tolerance to variations.
- Design Flexibility : Supports digital, analog, mixed-signal, memories and RF circuits.
- Wide Supply Range : Operates with 1V to >3.3V.
- Cost Effective : Mature technology and huge foundry support.
- Low Heat Dissipation : Suitable for portable devices.

Key Concept Visualization



Technical Illustrations



Key Point

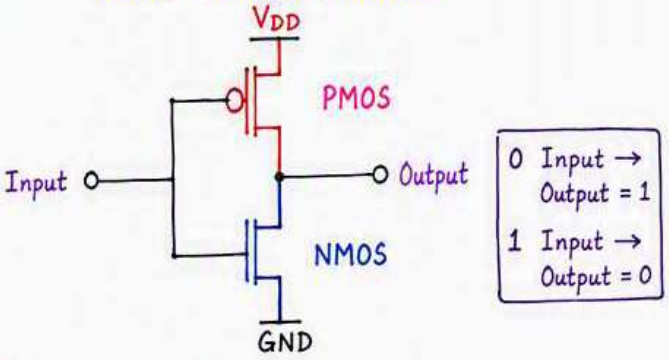
> CMOS is the dominant VLSI technology due to its ultra-low power, high density, excellent scalability and cost-effectiveness, making it the first choice for modern digital and mixed-signal ICs across all industries.

CMOS - An Industry Perspective

1. CMOS Technology Overview

- CMOS stands for Complementary Metal Oxide Semiconductor.
- Uses complementary NMOS and PMOS transistors.
- Forms the backbone of modern VLSI circuits.
- Widely used in digital IC design.
- Provides high integration density.
- Used in processors, memory and SoC devices.

Basic CMOS Inverter



Advantages of CMOS

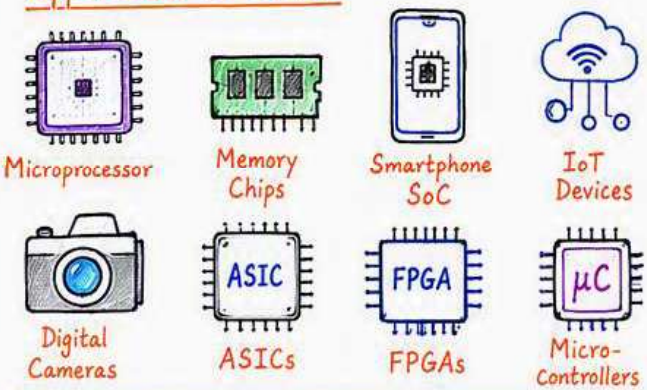
- ★ Very low static power consumption
- ★ High noise immunity
- ★ High input impedance
- ★ High packing density
- ★ Scalable for advanced nodes
- ★ Better power efficiency
- ★ High reliability



Disadvantages of CMOS

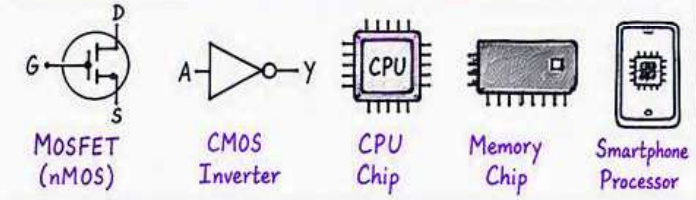
- ★ Sensitive to ESD damage
- ★ Fabrication process complexity
- ★ Latch-up possibility
- ★ Leakage current in advanced nodes
- ★ Performance trade-offs at very high speed

Applications of CMOS

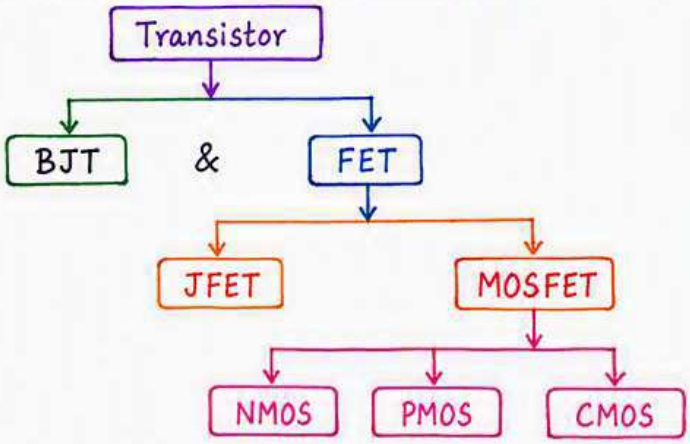


Industry Insight

CMOS dominates modern semiconductor manufacturing because it offers low power, high integration density, scalability and excellent reliability.

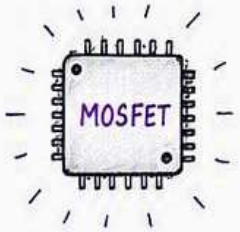


2. Classification of Transistors



MOSFET Applications

- ★ Digital ICs
- ★ CMOS Logic Gates
- ★ CPU Design
- ★ GPU Design
- ★ Memory Arrays
- ★ Low Power Electronics
- ★ Power Management ICs
- ★ VLSI Systems



3. BJT vs FET Comparison

Parameter	BJT	FET
Control Type	Current Controlled	Voltage Controlled
Input Impedance	Low	Very High
Power Consumption	Higher	Lower
Noise	More	Less
Speed	Moderate	High
Temperature Stability	Moderate	Better
Fabrication	More Complex	Suitable for ICs
Usage	Amplifiers	VLSI & CMOS Design

Key Difference : FET is voltage-controlled with very high input impedance and lower power.

Why FET is Preferred Over BJT?

- ★ Higher input impedance
- ★ Lower power consumption
- ★ Voltage-controlled device
- ★ Easy CMOS integration
- ★ Suitable for high-density ICs
- ★ Faster switching
- ★ Better scalability
- ★ Ideal for modern VLSI processes



Summary / Key Points

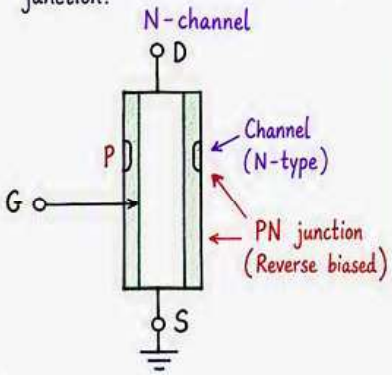
1. CMOS uses complementary NMOS and PMOS.
2. CMOS offers very low power consumption.
3. MOSFET is the basis of CMOS technology.
4. FET is voltage-controlled.
5. FET provides high input impedance.
6. CMOS dominates modern VLSI design.
7. High integration density is a key advantage.
8. Modern processors are CMOS based.

CMOS - An Industry Perspective

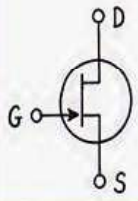
1. JFET (Junction Field Effect Transistor)

a) What is JFET?

JFET is a unipolar device in which current is controlled by the reverse biasing of the PN junction.

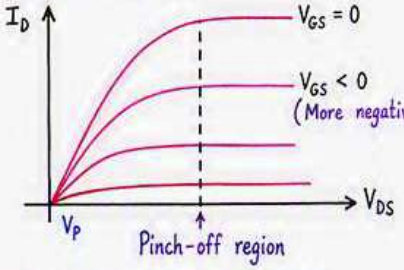


Symbol (N-channel)



b) Working Principle

- The gate is reverse biased w.r.t the channel.
- As V_{GS} becomes more negative, depletion regions widen and the channel narrows.
- When channel is narrow, I_D decreases.
- At pinch-off voltage (V_P), the channel is completely depleted and I_D becomes ~ 0 (cut-off).
- Current is conducted by majority carriers only (electrons in N-channel).



c) Advantages of JFET over BJT

- ★ Voltage controlled → zero gate current.
- ★ Very high input impedance ($10^7 - 10^9\ \Omega$).
- ★ Low noise and good thermal stability.
- ★ Lower power consumption.
- ★ Simple biasing and good linearity.



d) Disadvantages of JFET

- ★ Lower transconductance than BJT.
- ★ Limited drive current capability.
- ★ Gate junction can be damaged by excess reverse bias.
- ★ Not suitable for very high frequency (compared to MOSFET).
- ★ Larger device area for same performance.

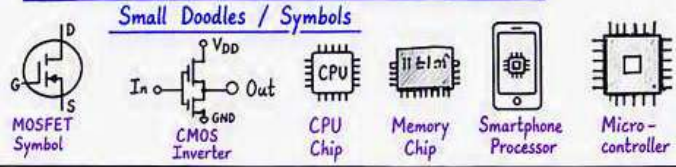


Key Takeaway (JFET)



JFET is a unipolar, high input impedance device, better than BJT for low power and low noise applications, but MOSFET is superior for integration and scaling.

Small Doodles / Symbols

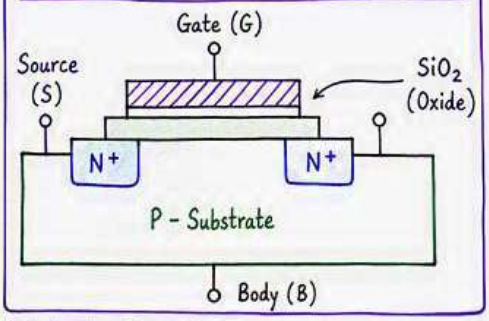


2. MOSFET (Metal Oxide Semiconductor FET)

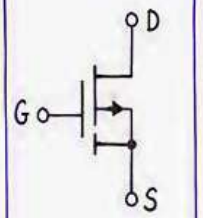
a) What is MOSFET?

MOSFET is a voltage controlled device in which the conduction path is formed/inverted by an electric field through an oxide insulated gate.

Structure (N-channel Enhancement MOSFET)

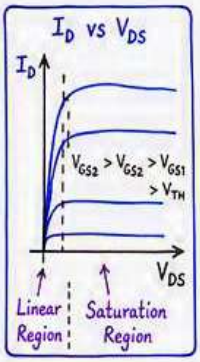


Symbol (N-MOSFET)



b) Working Principle

- With $V_{GS} = 0$, no conduction path exists between D & S.
- When $V_{GS} > V_{TH}$ (threshold voltage), an inversion channel is formed under the gate.
- Electrons flow from Drain to Source through this channel when V_{DS} is applied.
- Current is controlled by V_{GS} (voltage controlled device).



c) Advantages of MOSFET over JFET

- ★ Extremely high input impedance ($10^{10} - 10^{15}\ \Omega$).
- ★ Very low power consumption.
- ★ Better scalability (small dimensions).
- ★ Easier to integrate in ICs (CMOS technology).
- ★ Higher switching speed.
- ★ No reverse bias current at gate → no gate leakage (ideally).
- ★ Can be fabricated in complementary form (NMOS/PMOS).



Comparison Summary (JFET vs MOSFET)

Parameter	JFET	MOSFET
Gate Control	PN junction reverse bias	Electric field through oxide
Input Impedance	High ($10^7 - 10^9\ \Omega$)	Very high ($10^{10} - 10^{15}\ \Omega$)
Gate Current	~ 0 (but leakage possible)	~ 0 (ideally zero)
Power Consumption	Low	Very Low
Scalability	Moderate	Excellent
IC Integration	Difficult	Very Easy (CMOS)
Speed	Moderate	High
Best Use	Analog, Low noise	Digital, High density ICs

Industry Insight



MOSFET based CMOS technology is the backbone of modern semiconductors as it offers ultra high integration density, low power, high speed and excellent scalability.

Quick Revision Points

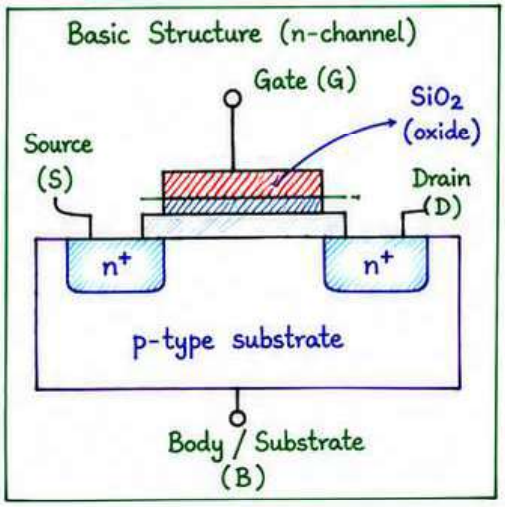
1. JFET → reverse biased PN junction, unipolar device.
2. JFET → high input impedance, low noise, low power.
3. JFET limitations → lower transconductance, not highly scalable.
4. MOSFET → voltage controlled, uses insulated gate (SiO_2).
5. MOSFET → extremely high input impedance & low power.
6. MOSFET → easy CMOS integration, highly scalable.
7. MOSFET is superior to JFET in almost all aspects.

CMOS – An Industry Perspective

1. MOSFET Introduction, Terminals, Features & Diagrams

Introduction :

MOSFET (Metal Oxide Semiconductor Field Effect Transistor) is a voltage controlled device. It uses an insulated gate (oxide) to control the flow of carriers between drain and source. It has very high input impedance, low power consumption and is the basic building block of VLSI circuits.

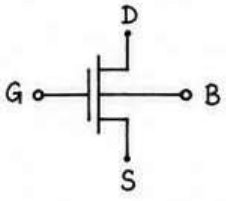


Terminals :

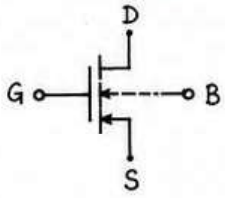
- Gate (G) : Controls channel formation / conduction.
- Drain (D) : Current enters or leaves the device.
- Source (S) : Reference terminal for carriers.
- Body (B) : Substrate terminal, usually connected to source in ICs.

Example Symbols (n-channel)

Enhancement type



Depletion type



Key Features :

- Very high input impedance (gate is insulated).
- Voltage controlled device.
- Low static power dissipation.
- Majority carrier device → fast switching.
- Scalable to very small dimensions.
- Used extensively in VLSI, digital & analog ICs.

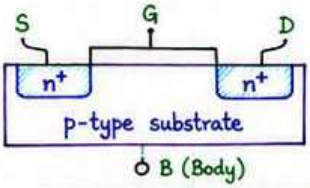
2. Types of MOSFET — Classification Based On :

MOSFET

A. Channel Type

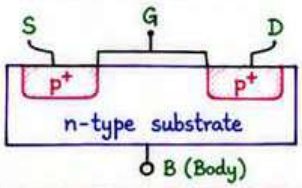
1. n-channel

- Electrons are the majority carriers.
- Built using p-type substrate with n^+ source & drain.
- Higher mobility → higher current drive.



2. p-channel

- Holes are the majority carriers.
- Built using n-type substrate with p^+ source & drain.
- Lower mobility → lower current drive.



B. Mode of Operation

i. Enhancement Mode

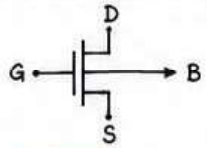
- No physical channel exists at $V_{GS} = 0$.
- Channel is induced by applying appropriate gate voltage.
- Device is normally OFF at $V_{GS} = 0$.

ii. Depletion Mode

- A physical channel exists even at $V_{GS} = 0$.
- Can be operated in both depletion and enhancement regions.
- Device is normally ON at $V_{GS} = 0$.

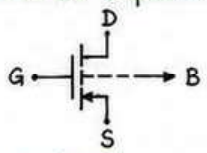
Combined Classification

1. NMOS Enhancement



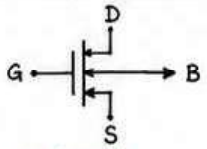
- n-channel
- Enhancement mode

2. NMOS Depletion



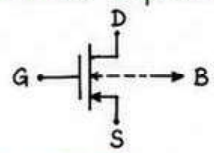
- n-channel
- Depletion mode

3. PMOS Enhancement



- p-channel
- Enhancement mode

4. PMOS Depletion



- p-channel
- Depletion mode

Note : Arrow in the symbol shows the direction of conventional current in the body diode. In ICs, body (B) is usually tied to source (S).

Author : Konda Tharun Vivekananda

CMOS – An Industry Perspective

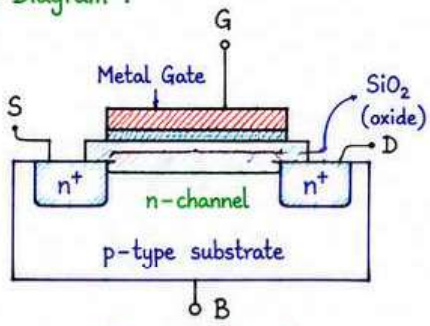
1. N channel Depletion MOSFET – operation, Diagram, definition, graphs – IV characteristics, Transient analysis

☑ **Definition :** An n-channel depletion MOSFET has a pre-formed n-type channel in the p-type body. The device conducts at $V_{GS} = 0$. Applying negative V_{GS} depletes the channel and can turn it OFF.

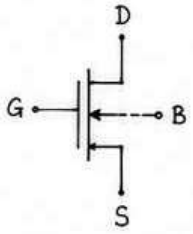
☑ **Operation :**

- At $V_{GS} = 0$, channel exists ⇒ current flows for $V_{DS} > 0$.
- For $V_{GS} < 0$, channel is depleted ⇒ I_D decreases.
- At $V_{GS} = V_{GS(off)}$ (negative), channel is cut off ⇒ $I_D \approx 0$.
- For $V_{GS} > 0$, channel is enhanced ⇒ I_D increases.

☑ **Diagram :**



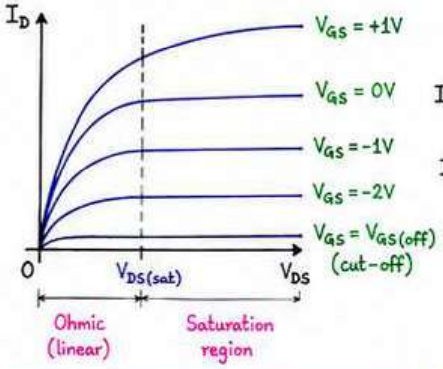
(Physical structure)

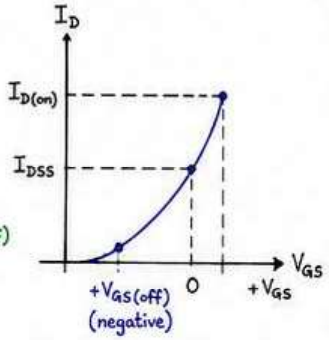


(Circuit symbol)

☑ **IV Characteristics :**

(a) Output Characteristics (I_D vs V_{DS}) (b) Transfer Characteristics (I_D vs V_{DS})

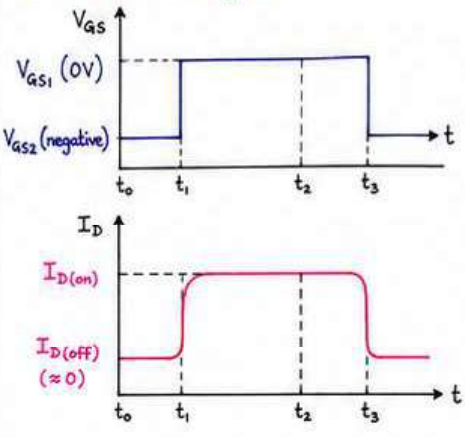




As V_{GS} becomes more negative ⇒ I_D decreases and at $V_{GS} = V_{GS(off)}$, device turns OFF.

At $V_{GS} = 0$, $I_D = I_{DSS}$ (finite). At $V_{GS} = V_{GS(off)}$ (negative), $I_D \approx 0$.

☑ **Transient Analysis :**



- When V_{GS} goes from 0 to negative, channel depletes and I_D falls.
- When V_{GS} returns to 0, channel recovers and I_D returns to I_{DSS}.
- If $V_{GS} = V_{GS(off)}$, I_D becomes ≈ 0.

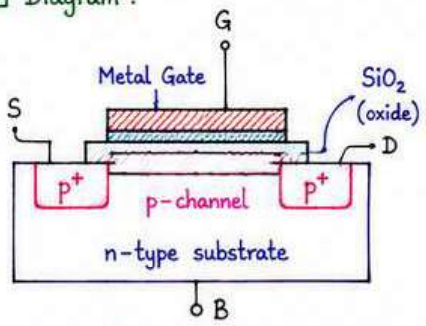
2. P channel Depletion MOSFET – operation, Diagram, definition, graphs – IV characteristics, Transient analysis

☑ **Definition :** A p-channel depletion MOSFET has a pre-formed p-type channel in the n-type body. The device conducts at $V_{GS} = 0$. Applying positive V_{GS} depletes the channel and can turn it OFF.

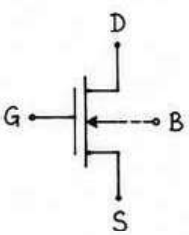
☑ **Operation :**

- At $V_{GS} = 0$, channel exists ⇒ current flows for $V_{DS} < 0$.
- For $V_{GS} > 0$, channel is depleted ⇒ I_D decreases.
- At $V_{GS} = V_{GS(off)}$ (positive), channel is cut off ⇒ $I_D \approx 0$.
- For $V_{GS} < 0$, channel is enhanced ⇒ I_D increases (in magnitude).

☑ **Diagram :**



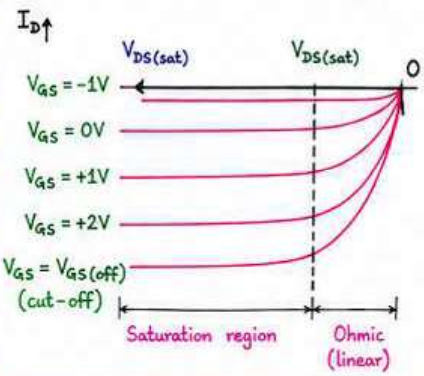
(Physical structure)

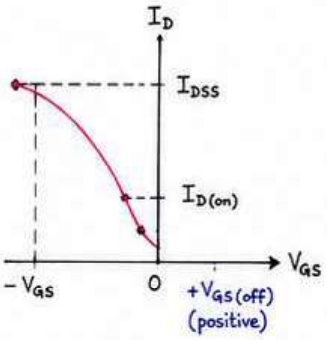


(Circuit symbol)

☑ **IV Characteristics :**

(a) Output Characteristics (I_D vs V_{DS}) (b) Transfer Characteristics (I_D vs V_{GS})

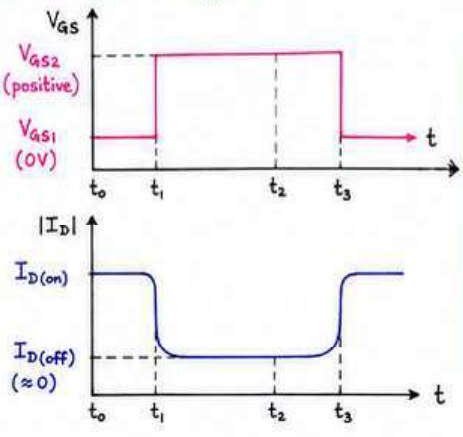




As V_{GS} becomes more positive ⇒ I_D decreases and at $V_{GS} = V_{GS(off)}$, device turns OFF.

At $V_{GS} = 0$, $I_D = I_{DSS}$ (finite). At $V_{GS} = V_{GS(off)}$ (positive), $I_D \approx 0$.

☑ **Transient Analysis :**



- When V_{GS} goes from 0 to positive, channel depletes and $|I_D|$ falls.
- When V_{GS} returns to 0, channel recovers and $|I_D|$ returns to I_{DSS}.
- If $V_{GS} = V_{GS(off)}$, $|I_D|$ becomes ≈ 0.

Author : Konda Tharun Vivekananda

CMOS – An Industry Perspective

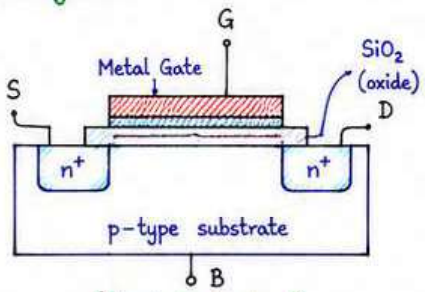
1. N channel Enhancement MOSFET – operation, Diagram, definition, graphs – IV characteristics, Transient analysis

☑ **Definition :** An n-channel enhancement MOSFET has no physical channel at $V_{GS} = 0$. A conducting channel is induced in the p-type body when $V_{GS} > V_T$ (positive). Below V_T the device is OFF.

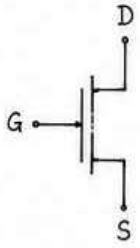
☑ **Operation :**

- At $V_{GS} < V_T$ (positive), no channel ⇒ $I_D \approx 0$.
- At $V_{GS} = V_T$, inversion channel starts forming ⇒ I_D starts to flow (small).
- At $V_{GS} > V_T$, channel strength increases ⇒ I_D increases.
- Increasing V_{DS} increases I_D until saturation.

☑ **Diagram :**



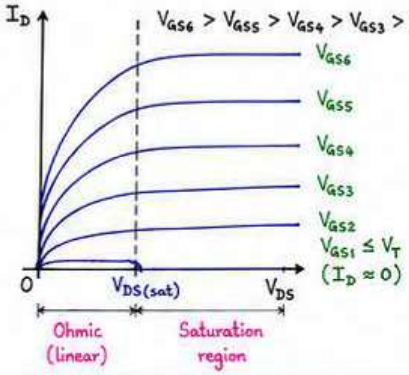
(Physical structure)

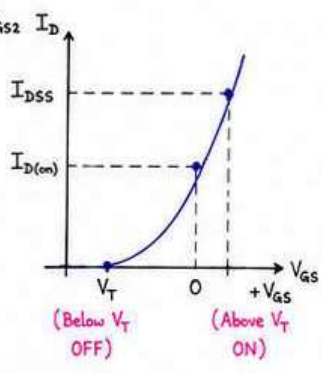


(Circuit symbol)

☑ **IV Characteristics :**

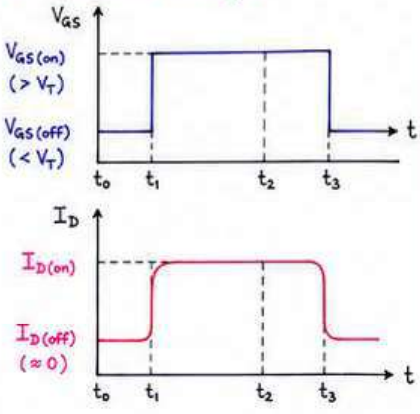
(a) Output Characteristics (I_D vs V_{DS}) (b) Transfer Characteristics (I_D vs V_{GS})





- For $V_{GS} < V_T \Rightarrow I_D \approx 0$ (device OFF).
- As V_{GS} increases above V_T, I_D increases.
- For large V_{DS}, I_D saturates at I_{DSS}.

☑ **Transient Analysis :**



- When V_{GS} steps from below V_T to above V_T, channel forms and I_D rises.
- When V_{GS} falls below V_T, channel disappears and I_D returns to ≈ 0.
- Above V_T, I_D depends on V_{GS} and V_{DS}.

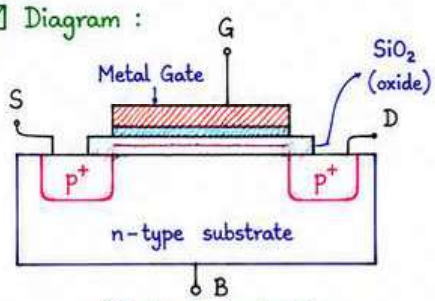
2. P channel Enhancement MOSFET – operation, Diagram, definition, graphs – IV characteristics, Transient analysis

☑ **Definition :** A p-channel enhancement MOSFET has no physical channel at $V_{GS} = 0$. A conducting channel is induced in the n-type body when $V_{GS} < -|V_T|$ (negative). Above $-|V_T|$ the device is OFF.

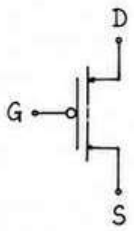
☑ **Operation :**

- At $V_{GS} > -|V_T|$ (less negative), no channel ⇒ $I_D \approx 0$.
- At $V_{GS} = -|V_T|$, inversion channel starts forming ⇒ I_D starts to flow (small).
- At $V_{GS} < -|V_T|$ (more negative), channel strength increases ⇒ $|I_D|$ increases.
- Increasing $|V_{DS}|$ increases $|I_D|$ until saturation.

☑ **Diagram :**



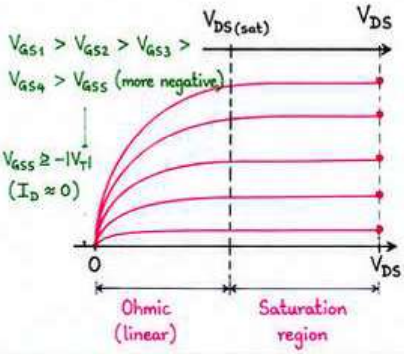
(Physical structure)

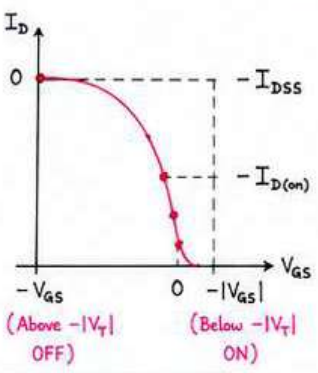


(Circuit symbol)

☑ **IV Characteristics :**

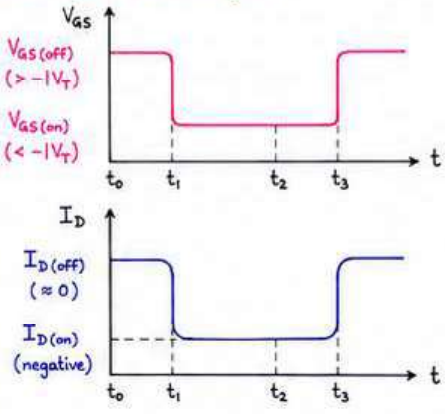
(a) Output Characteristics (I_D vs V_{DS}) (b) Transfer Characteristics (I_D vs V_{GS})





- For $V_{GS} > -|V_T| \Rightarrow I_D \approx 0$ (device OFF).
- As V_{GS} becomes more negative than $-|V_T|$, $|I_D|$ increases.
- For large $|V_{DS}|$, $|I_D|$ saturates at I_{DSS}.

☑ **Transient Analysis :**



- When V_{GS} goes from above $-|V_T|$ to below $-|V_T|$, channel forms and I_D becomes negative.
- When V_{GS} goes above $-|V_T|$, channel disappears and I_D returns to ≈ 0.
- Below $-|V_T|$, $|I_D|$ depends on V_{GS} and $|V_{DS}|$.

Author : Konda Tharun Vivekananda

CMOS – An Industry Perspective

1. Threshold Voltage and its importance

☑ Definition :

Threshold voltage (V_T) is the minimum gate-to-source voltage required to form an inversion layer (strong channel) in a MOSFET.

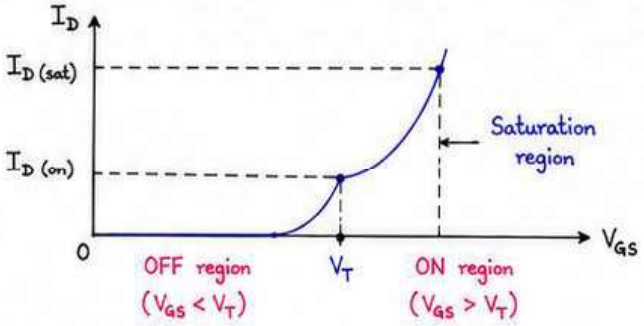
Below V_T → device is OFF.

Above V_T → device begins to conduct.

☑ How V_T is defined ? (nMOS example)

- In the transfer characteristic (I_D vs V_{GS}) in saturation region.
- V_T is the value of V_{GS} at which I_D starts increasing rapidly (knee of the curve).
- Practically, I_D = small current (e.g., 10^{-7} A·W/L) is used to extract V_T.

☑ nMOS Transfer Characteristic (I_D vs V_{GS})



☑ Importance of Threshold Voltage

- Controls the turn-ON/turn-OFF of MOSFET.
- Determines the switching speed (lower V_T → faster).
- Affects leakage current (lower V_T → higher leakage).
- Impacts power consumption in CMOS circuits.
- Critical parameter in device scaling and low-power design.

☑ Typical V_T Values (at Room Temperature)

Process / Node (approx)	nMOS V_T (V)	pMOS $\lvert V_T \rvert$ (V)
180 nm	0.40 – 0.50	0.40 – 0.50
90 nm	0.25 – 0.35	0.25 – 0.35
45 nm	0.15 – 0.25	0.15 – 0.25
22 nm and below	0.10 – 0.20	0.10 – 0.20

☑ Key Points

- V_T depends on doping, oxide thickness, material, and process.
- Lower $\lvert V_T \rvert$ improves speed but increases leakage.
- Proper V_T selection balances performance and power.

2. Applications of CMOS

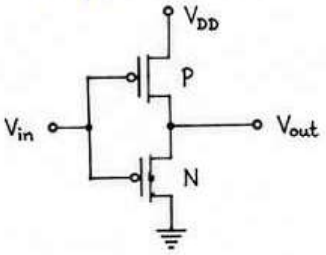
CMOS is the most widely used technology in VLSI and digital/analog ICs due to its low power, high noise margin and high integration capability.

☑ Major Applications

1. Digital Logic Circuits

- CMOS is used to implement logic gates, multiplexers, decoders, adders, etc.
- Basis for all digital ICs.

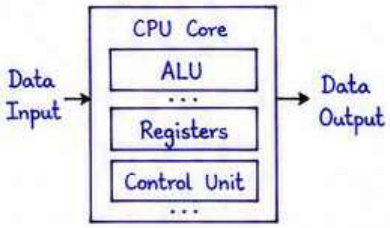
Example : CMOS Inverter



2. Microprocessors and CPUs

- Core building blocks (logic, registers, ALUs) are implemented using CMOS.
- Used in general-purpose processors, SoCs.

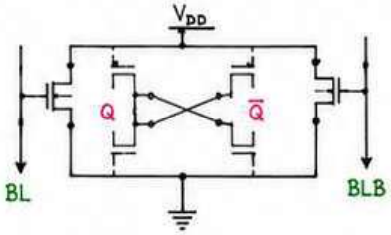
Example : CPU Core



3. Memory Circuits

- SRAM cells, DRAM cells, ROM, Flash memory are implemented using CMOS.
- High density and low power operation.

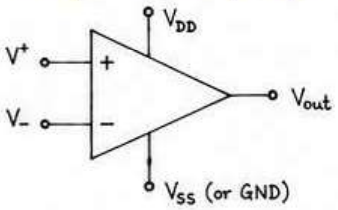
Example : 6T SRAM Cell



4. Analog and Mixed Signal ICs

- CMOS used in amplifiers, comparators, ADCs, DACs, PLL, references.
- Low power analog/mixed signal applications.

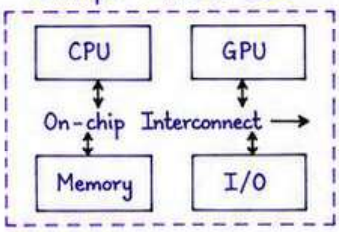
Example : CMOS Op-Amp



5. Low Power and Portable Devices

- Used in smartphones, tablets, wearables, IoT devices.
- CMOS offers ultra-low power in standby and active modes.

Example : Mobile SoC



☑ Key Advantages in Applications

- Very low static power (ideally zero).
- High noise immunity and full voltage swing.
- High packing density and scalability.
- Compatible with both digital and analog design.

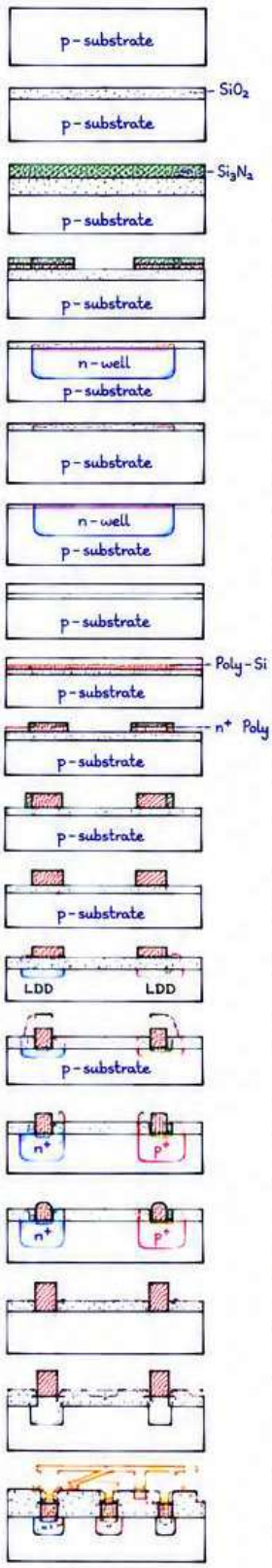
Author : Konda Tharun Vivekananda

CMOS – An Industry Perspective

1. CMOS Fabrication using N well

☑ Process Flow : (19 Stages)

1. Start with p-type silicon wafer (p-substrate).
2. Grow thin pad oxide (SiO_2).
3. Deposit Si_3N_4 layer (mask for well).
4. Photolithography to open N-well region.
5. Implant / diffuse Phosphorus to form N-well.
6. Drive-in / anneal to form N-well.
7. Strip Si_3N_4 and pad oxide.
8. Grow gate oxide (thin SiO_2).
9. Deposit polysilicon layer.
10. Dope poly-Si (n^+).
11. Photolithography to pattern poly gates.
12. Etch polysilicon to form gate electrodes.
13. Lightly implant (LDD) for NMOS and PMOS.
14. Deposit spacer oxide (SiO_2) and etch spacers.
15. Perform source / drain implant :
    - NMOS : n^+ implant (As / P)
    - PMOS : p^+ implant (B)
16. Anneal to activate dopants.
17. Deposit interlayer dielectric (ILD, SiO_2).
18. Open contact holes (to S/D and gates).
19. Metallization (Al / Cu) and pattern interconnects. (Then passivation layer if required).



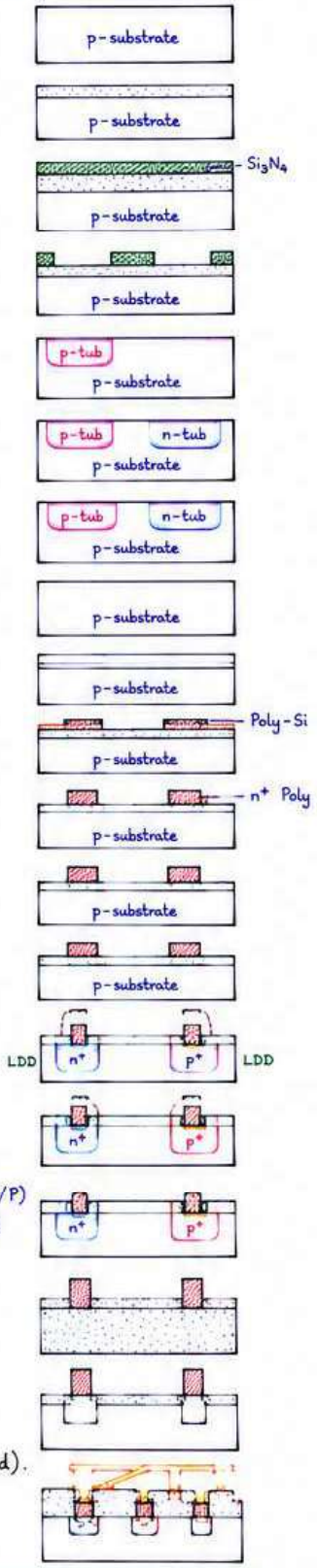
☑ Key Points :

- N-well is used to build PMOS in a p-type substrate.
- NMOS is built directly in the p-substrate.
- Isolation between devices is by reverse-biased junctions.
- Suitable for low to medium density processes.

2. CMOS Fabrication using Twin Tub

☑ Process Flow : (19 Stages)

1. Start with p-type silicon wafer (p-substrate).
2. Grow thin pad oxide (SiO_2).
3. Deposit Si_3N_4 layer (mask for tubs).
4. Photolithography to open P-tub and N-tub regions.
5. Implant Boron to form P-tub.
6. Implant Phosphorus to form N-tub.
7. Drive-in / anneal to form tubs.
8. Strip Si_3N_4 and pad oxide.
9. Grow gate oxide (thin SiO_2).
10. Deposit polysilicon layer.
11. Dope poly-Si (n^+).
12. Photolithography to pattern poly gates.
13. Etch polysilicon to form gate electrodes.
14. Lightly implant (LDD) for NMOS and PMOS.
15. Deposit spacer oxide (SiO_2) and etch spacers.
16. Perform source / drain implant :
    - NMOS (in N-tub) : n^+ implant (As/P)
    - PMOS (in P-tub) : p^+ implant (B)
17. Anneal to activate dopants.
18. Deposit interlayer dielectric (ILD, SiO_2).
19. Open contact holes and Metallization (Al / Cu). (Then passivation layer if required).



☑ Key Points :

- Both PMOS (in P-tub) and NMOS (in N-tub) are formed in the same p-substrate.
- Better isolation and less latch-up tendency.
- Suited for high performance and high density CMOS.
- Requires one extra mask and two implants.

p-substrate | n-well / n-tub | p-tub | SiO_2 (oxide) | Si_3N_4 (nitride) | Poly-Si | Metal (Al/Cu)

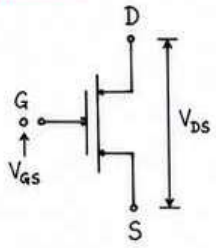
Author : Konda Tharun Vivekananda

CMOS – An Industry Perspective

1. MOSFET Operating Regions – Cut off, Linear, Saturation and behaviour and graphs

☑ General (nMOS enhancement type)

- V_{GS} ⇒ Gate to Source voltage
- V_{DS} → Drain to Source voltage
- V_T → Threshold voltage
- Regions depend on V_{GS} and V_{DS}.



Region conditions (nMOS)

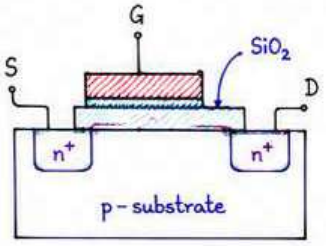
Region	Conditions
Cut off	$V_{GS} \le V_T$
Linear (Triode)	$V_{GS} > V_T$ and $0 \le V_{DS} < V_{GS} - V_T$
Saturation	$V_{GS} > V_T$ and $V_{DS} \ge V_{GS} - V_T$

(pMOS : Replace n with p and reverse voltage polarities.)

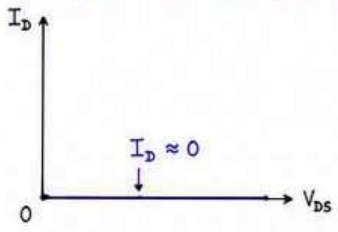
1. Cut off Region

☑ Condition : $V_{GS} \le V_T$

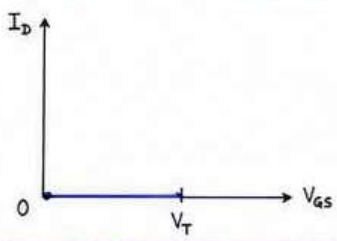
- No inversion channel is formed.
- $I_D \approx 0$ (only leakage current).
- Device is OFF.



$I_D - V_{DS}$ Characteristics ($V_{GS} \le V_T$)



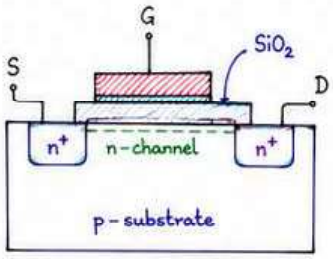
$I_D - V_{GS}$ Characteristics ($V_{DS} > 0$)



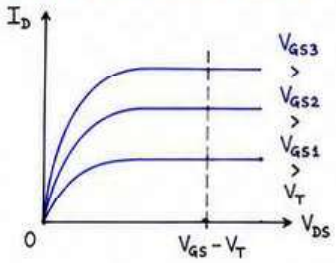
2. Linear (Triode) Region

☑ Condition : $V_{GS} > V_T$ and $0 \le V_{DS} < V_{GS} - V_T$

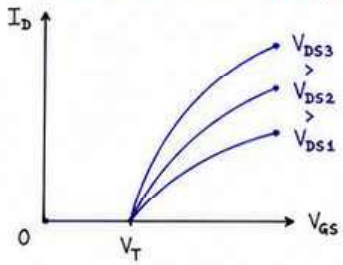
- Inversion channel exists along entire length.
- I_D increases with V_{DS}.
- Behaves like a voltage-controlled resistor.



$I_D - V_{DS}$ Characteristics (Linear)



$I_D - V_{GS}$ Characteristics (Linear)



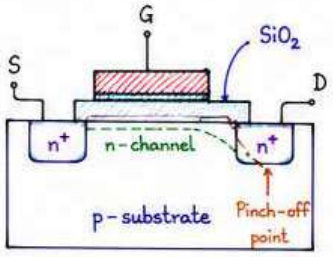
$$I_D = \mu_n C_{ox} \left(\frac{W}{L}\right)\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right] \quad (0 \le V_{DS} < V_{GS} - V_T)$$

where μ_n : mobility of electrons
C_{ox} : gate oxide capacitance / unit area
W/L : width to length ratio

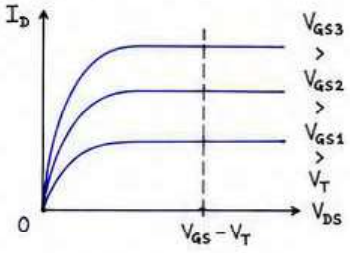
3. Saturation Region

☑ Condition : $V_{GS} > V_T$ and $V_{DS} \ge V_{GS} - V_T$

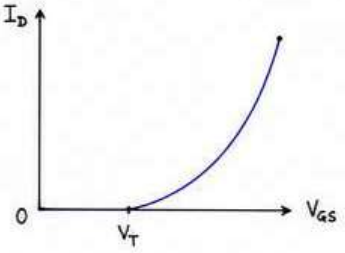
- Channel exists with pinch-off near drain.
- I_D becomes almost constant (independent of V_{DS}).
- Used for amplification.



$I_D - V_{DS}$ Characteristics (Saturation)



$I_D - V_{GS}$ Characteristics (Saturation)



$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_T)^2(1 + \lambda V_{DS}) \quad (V_{DS} \ge V_{GS} - V_T)$$

λ : channel length modulation parameter
($\lambda \approx 0$ for ideal case)

4. Summary of Regions

Region	Condition	Channel Status	I_D Behaviour	Use / Behaviour
Cut off	$V_{GS} \le V_T$	No inversion channel	$I_D \approx 0$	Switch OFF
Linear (Triode)	$0 \le V_{DS} < V_{GS} - V_T$	Channel along full length	$I_D \propto V_{DS}$	Variable resistor / Low V_{DS} operation
Saturation	$V_{DS} \ge V_{GS} - V_T$	Channel pinch-off near drain	$I_D \approx$ constant	Amplification / Current source

☑ Note : For pMOS, replace n with p. Voltage polarities are reversed.

pMOS Conditions :

- Cut off : $V_{SG} \le |V_T|$
- Linear : $V_{SG} > |V_T|$ and $0 \le V_{SD} < V_{SG} - |V_T|$
- Saturation : $V_{SG} > |V_T|$ and $V_{SD} \ge V_{SG} - |V_T|$

Author : Konda Tharun Vivekananda

CMOS – An Industry Perspective

1. MOSFET Channel Effects – Short Channel Effects

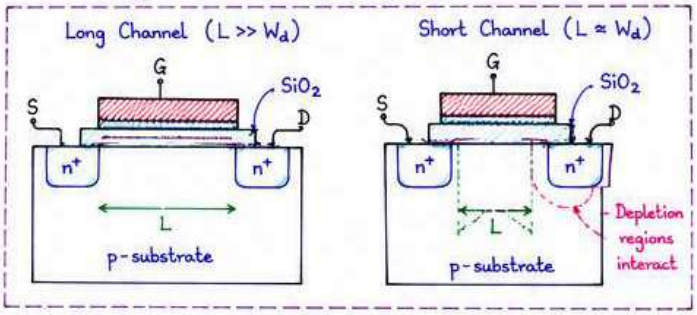
☑ Introduction :

When the channel length (L) becomes comparable to the depletion region width at high V_{DS}, the ideal long channel assumptions break down and several short channel effects (SCEs) appear, degrading device performance.

☑ Why Short Channel Effects Occur ?

- Drain electric field penetrates into the channel.
- Source and drain depletion regions start interacting.

Long Channel vs Short Channel



☑ Major Short Channel Effects (SCEs)

1. Threshold Voltage Roll-off – Threshold voltage decreases.
2. Drain Induced Barrier Lowering (DIBL) – Barrier at source reduces due to drain voltage.
3. Subthreshold Slope Degradation – Slope (mV/decade) increases.
4. Punchthrough – Channel is controlled by drain.
5. Velocity Saturation – Carrier velocity saturates.
6. Hot Carrier Effects – High energy carriers cause damage.
7. Increased Off-state Leakage – Higher I_{off}.
8. Mobility Degradation – Due to vertical field.

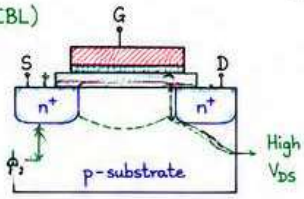
☑ 1. Threshold Voltage Roll-off

- V_T decreases as channel length decreases.
- Caused by control from drain electrical field.



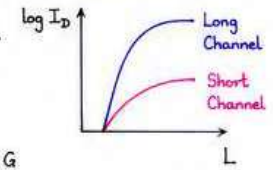
☑ 2. Drain Induced Barrier Lowering (DIBL)

- Increase in V_{DS} lowers the potential barrier at source.
- Leads to lower V_T and higher off current.



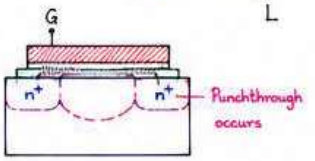
3. Subthreshold Slope Degradation

- Slope (S) increases from ideal 60 mV/dec.
- Poor gate control due to charge sharing.



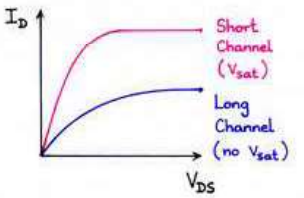
4. Punchthrough

- Depletion regions from source and drain meet even at low V_{DS}.
- Drain controls channel current.



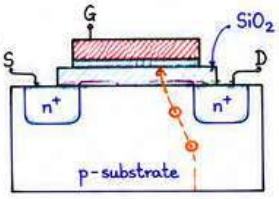
5. Velocity Saturation

- At high vertical field, carriers reach saturation velocity (v_{sat}).
- I_D increases slowly with V_{DS}.



6. Hot Carrier Effects

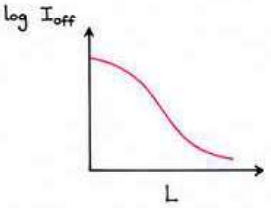
- High energy carriers (from high V_{DS}) get trapped in oxide.
- Causes threshold shift and reliability issues.



Hot carriers get trapped in oxide

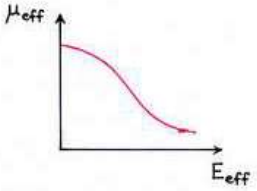
7. Increased Off-state Leakage

- Due to DIBL and punchthrough.
- I_{off} increases exponentially.



8. Mobility Degradation

- High vertical field and surface scattering reduce mobility.
- Lower I_{on} and transconductance.



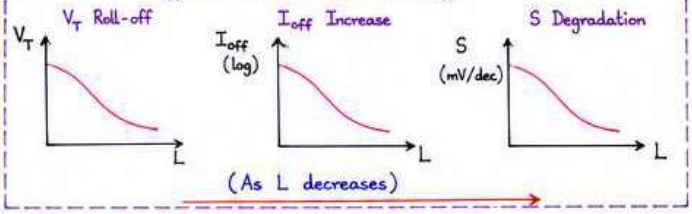
Effect Summary

S.No.	Effect	Cause	Impact
1	Threshold Voltage Roll-off	Drain field penetration	Lower V_T Higher leakage
2	DIBL	Drain field lowers source barrier	Lower V_T Higher I_{off}
3	Subthreshold Slope Degradation	Poor gate control	Higher S (mV/dec)
4	Punchthrough	Depletion regions meet	Loss of gate control
5	Velocity Saturation	High vertical field	Lower drive current
6	Hot Carrier Effects	High energy carriers	Reliability issues
7	Increased Off Leakage	DIBL + Punchthrough	Higher I_{off}.
8	Mobility Degradation	High field + scattering	Lower μ_{eff}, Lower g_m

☑ Scaling Impact

- As L scales down (< 100 nm), short channel effects become severe.
- Design challenges : leakage, power, reliability, variability.
- Solutions : Halo implant, Pocket implant, High-k gate dielectric, Metal gate, Strained silicon, UTB / FinFET.
- Trade-off between performance, power and reliability.

Typical Trend with Channel Length



☑ Key Takeaway

Short channel effects are inevitable in scaled MOSFETs. Understanding them is essential for designing high performance, low power and reliable VLSI circuits.

Author : Konda Tharun Vivekananda

CMOS – An Industry Perspective

1. MOSFET Channel Effects – Narrow Channel Effects

☑ Introduction :

When the channel width (W) becomes comparable to the depletion region widths from the source and drain, the device behavior deviates from ideal long channel operation. These are called Narrow Channel Effects.

☑ Why Narrow Channel Effects Occur ?

- Lateral depletion regions from S and D extend in the sidewalls.
- Effective channel width gets reduced.
- Gate controls the channel less effectively.
- More pronounced as W decreases.

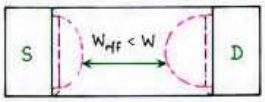
Cross-section (Top view)

Wide Channel ($W >> X_d$)



Depletion regions do not overlap

Narrow Channel ($W \approx 2X_d$)



Depletion regions overlap

☑ Major Narrow Channel Effects

1. Narrow width effect (effective width reduction)
2. Threshold voltage roll-off in narrow width
3. Increased off-state leakage
4. Drain induced barrier lowering (DIBL) enhancement
5. Mobility degradation due to sidewall scattering
6. Velocity saturation enhancement
7. Punchthrough in narrow width
8. Hot carrier effects enhancement
9. Transconductance reduction
10. Output resistance reduction

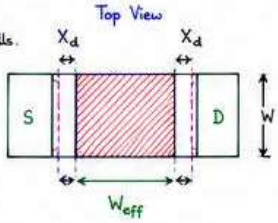
W = Physical channel width
W_{eff} = Effective channel width
X_d = Lateral depletion width

1. Narrow Width Effect (Effective Width Reduction)

- Depletion regions from S and D encroach from sidewalls.
- Effective width is reduced.

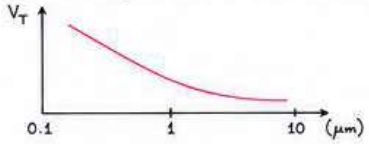
$$W_{eff} = W - 2X_d \quad (\text{for } W > 2X_d)$$

Top View



2. Threshold Voltage Roll-off in Narrow Width

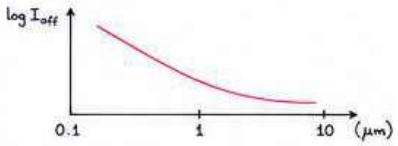
- Due to charge sharing and sidewall depletion.
- Threshold voltage reduces as width decreases.



V_T decreases as width decreases

3. Increased Off-state Leakage

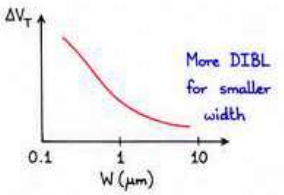
- Overlap of depletion regions reduces channel control.
- Larger subthreshold leakage and junction leakage.
- I_{off} increases as W decreases.



I_{off} increases as width decreases

4. DIBL Enhancement

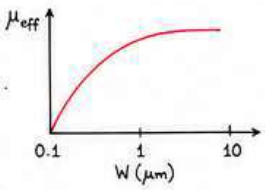
- Drain field penetrates more in narrow width.
- Barrier lowering at source is more.
- Higher off current and lower V_T.



More DIBL for smaller width

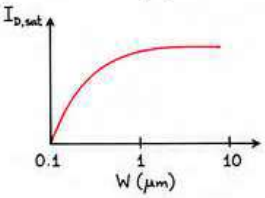
5. Mobility Degradation (Sidewall Scattering)

- Carriers collide with sidewall roughness.
- Mobility (μ_{eff}) reduces as width decreases.



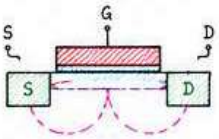
6. Velocity Saturation Enhancement

- Strong lateral electric field near sidewalls.
- Earlier velocity saturation.
- Reduces drive current.



7. Punchthrough in Narrow Width

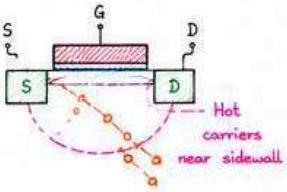
- Depletion regions from S and D meet even at lower V_{DS}.
- Channel is depleted.
- Device turns off prematurely.



Punchthrough occurs

8. Hot Carrier Effects Enhancement

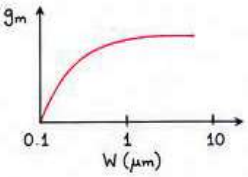
- High lateral field near drain sidewall.
- More hot carriers get generated.
- Reliability issues (oxide trapped charge).



Hot carriers near sidewall

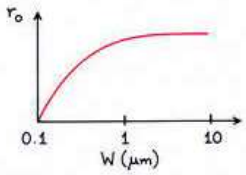
9. Transconductance Reduction (g_m)

- Due to mobility reduction and charge sharing.
- g_m reduces as width decreases.



10. Output Resistance Reduction (r_o)

- Due to enhanced DIBL and channel length modulation.
- r_o reduces with decreasing width.



Summary Table

S.No.	Effect	Cause	Impact
1	Narrow Width Effect	Sidewall depletion encroachment	$W_{eff} < W$
2	V_T Roll-off	Charge sharing, sidewall depletion	Lower V_T
3	Increased Off Leakage	Poor gate control, overlap of depletion	Higher I_{off}
4	DIBL	Lateral drain field penetration	Lower V_T, Higher I_{off}
5	Mobility Degradation	Sidewall scattering	Lower μ_{eff}
6	Velocity Saturation Enhancement	Strong lateral field	Lower $I_{D,sat}$
7	Punchthrough	Depletion regions meet early	Early turn-off
8	Hot Carrier Enhancement	High lateral field near drain	Reliability degradation
9	g_m Reduction	μ reduction + charge sharing	Lower g_m
10	r_o Reduction	Enhanced DIBL & channel modulation	Lower r_o

☑ Key Takeaway

- Narrow channel effects become significant when $W \approx 2X_d$.
- They degrade performance: lower drive current, higher leakage, lower V_T, and poor reliability.
- Careful device design and scaling techniques are essential.

To minimize:
- Increase W
- Use Halo implants
- Use LDD
- Optimize layout

Author : Konda Tharun Vivekananda

CMOS – An Industry Perspective

1. MOSFET Effects – Second Order Effects

☑ Introduction :

After first order phenomena, the following second order effects also influence MOSFET behaviour, especially as dimensions scale.

☑ List of Second Order Effects

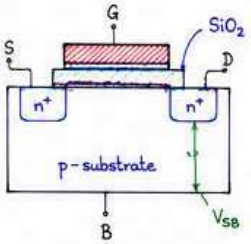
1. Body Effect (Substrate Bias Effect)
2. Channel Length Modulation.
3. Mobility Degradation (Vertical Field Effect)
4. Velocity Saturation
5. Drain Induced Barrier Lowering (DIBL)
6. Source Induced Barrier Lowering
7. Punchthrough in Short Channel
8. Hot Carrier Injection
9. GIDL (Gate Induced Drain Leakage)
10. Gate Leakage / Tunneling
11. Quantum Confinement (Narrow Body Effect)
12. Poly Depletion / Gate Resistance Effect

1. Body Effect (Substrate Bias Effect)

- Threshold voltage increases with reverse body bias.
- Decreases with forward body bias.

$$V_T = V_{T0} + \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right)$$

$$V_{SB} = V_S - V_B$$

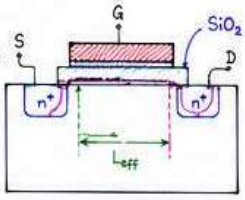


2. Channel Length Modulation

- Effective channel length reduces in saturation due to drain voltage increase.
- Causes increase in I_D with V_{DS}.

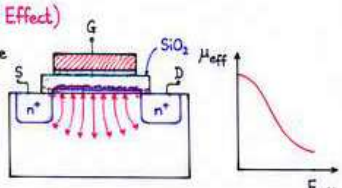
$$I_D = \frac{1}{2}\mu_n C_{ox}\left[\frac{W}{L}\right](V_{GS} - V_T)^2(1 + \lambda V_{DS})$$

λ : channel length modulation parameter



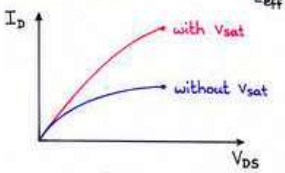
3. Mobility Degradation (Vertical Field Effect)

- High vertical electric field near the interface reduces mobility.
- Results in lower drive current.



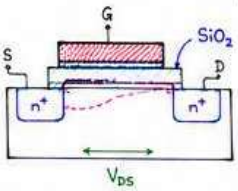
4. Velocity Saturation

- At high lateral field, carrier velocity saturates to v_{sat}.
- I_D increases linearly with V_{DS} instead of quadratically.



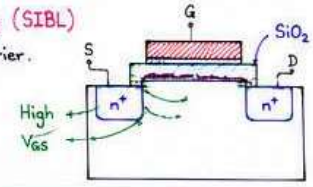
5. Drain Induced Barrier Lowering (DIBL)

- High V_{DS} lowers the barrier at source end.
- Lowers V_T and increases off current.



6. Source Induced Barrier Lowering (SIBL)

- High V_{GS} lowers the source barrier.
- Increases subthreshold current.

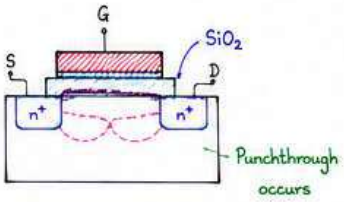


☑ Key Points :

- Second order effects become significant in deep submicron MOSFETs.
- They degrade performance, increase leakage and affect reliability.
- Careful device design and advanced fabrication techniques are essential.

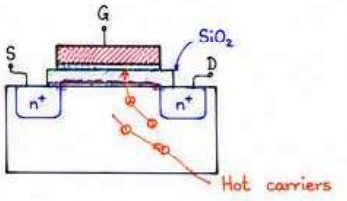
7. Punchthrough in Short Channel

- Depletion regions from source and drain touch each other.
- Channel disappears even at low V_{DS}.
- Device turns OFF.



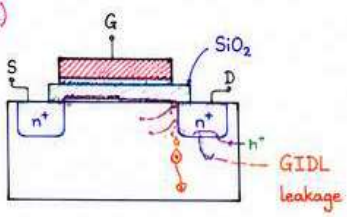
8. Hot Carrier Injection

- High energy carriers injected into oxide.
- Causes threshold shift, mobility degradation and reliability issues.



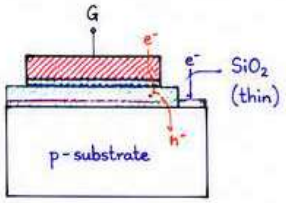
9. GIDL (Gate Induced Drain Leakage)

- At high V_{DS}, high field near drain causes band to band tunneling.
- Leakage from gate to drain.



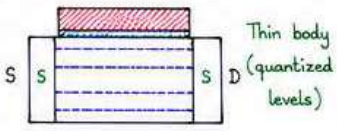
10. Gate Leakage / Tunneling

- Very thin oxide leads to direct electron tunneling.
- Increases gate leakage current I_G.



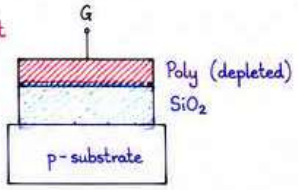
11. Quantum Confinement (Narrow Body Effect)

- When body thickness is comparable to de-Broglie wavelength.
- Energy levels become quantized.
- Increases V_T and reduces mobility.



12. Poly Depletion / Gate Resistance Effect

- Gate poly gets depleted.
- Increases gate resistance.
- Reduces transconductance g_m.



Summary Table

S.No.	Effect	Cause	Impact
1	Body Effect	Reverse body bias	↑ V_T, ↓ I_D
2	Channel Length Modulation	V_{DS} ↑ reduces L_{eff}	↑ I_D in saturation
3	Mobility Degradation	High vertical field	↓ μ_{eff}, ↓ I_D
4	Velocity Saturation	High lateral field	Linear I_D vs V_{DS}
5	DIBL	High V_{DS}	↓ V_T, ↑ off current
6	SIBL	High V_{GS}	↑ subthreshold current
7	Punchthrough	Short channel	Loss of control
8	Hot Carrier Injection	High energy carriers	Reliability degradation
9	GIDL	High field near drain	Gate to drain leakage
10	Gate Leakage	Thin oxide tunneling	↑ I_G
11	Quantum Confinement	Narrow body	↑ V_T, ↓ μ_{eff}
12	Poly Depletion	Poly gets depleted	↑ R_G, ↓ g_m

Design Considerations

- Use LDD, Halo implant, Pocket implant to control SCEs.
- Use high-k gate dielectric and metal gate to reduce leakage.
- Optimize channel length, width and body thickness.

Author : Konda Tharun Vivekananda

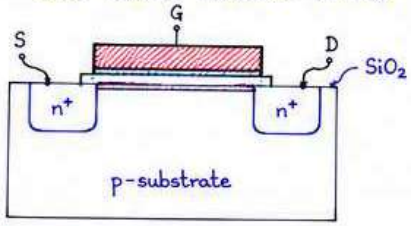
CMOS – An Industry Perspective

1. MOSFET Capacitance – introduction, diagrams, everything including oxide capacitance Complete concepts

☑ Introduction :

- Capacitances in MOSFET arise due to charges stored in oxide and depletion / inversion regions.
- They are bias dependent and important in AC / high speed behaviour.
- Total gate capacitance = oxide capacitance in series/parallel with depletion or inversion capacitances.

Basic MOSFET Structure (nMOS)



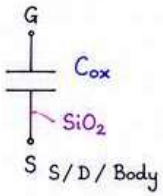
2 Oxide Capacitance (C_{ox})

- Capacitance of gate oxide per unit area.
- Independent of bias (ideal case).

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \quad (F/cm^2)$$

where $\varepsilon_{ox} = \varepsilon_0 \varepsilon_{oxr}$, t_{ox} = oxide thickness

$\varepsilon_0 = 8.854 \times 10^{-14}$ F/cm, $\varepsilon_{oxr} \approx 3.9$ (SiO_2)



3 Depletion Capacitances

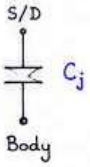
- Due to depletion region in the semiconductor.
- Depends on bias (V_G, V_D, V_S, V_B).
- Two types : C_{js} (source junction), C_{jd} (drain junction) and C_b (body or substrate capacitance).

(a) Source / Drain Junction Capacitance (C_J)

- Reverse biased p-n junction capacitance.
- Per unit area :

$$C_j = \frac{C_{jo}}{(1 + V_R/\phi_o)^m}$$

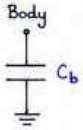
where C_{jo} = zero bias junction cap.
V_R = reverse bias voltage
ϕ_o = built-in potential (~0.6 – 0.8V)
m = grading coefficient (0.33 for abrupt, 0.5 for linearly graded)



(b) Body Capacitance (C_b)

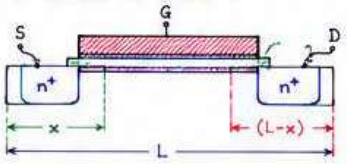
- Between body and ground (or bulk).

$C_b = C_{js} + C_{jD}$



4 Capacitance Partitioning (Ward-Dutton Model)

- Gate capacitance is split between source and drain depending on channel position.
- Charge at an inversion sheet is divided proportionally to distance to S and D.



Fractions :
To Source = x/L
To Drain = (L-x)/L

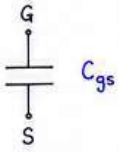
Capacitance Symbols & Meaning

Symbol	Meaning	Units
C_{ox}	Oxide capacitance per unit area	F/cm^2
C_{oxp}	Oxide fringe capacitance per unit length	F/cm
C_{js}, C_{jD}	Source/Drain junction capacitances	F
C_b	Body (substrate) capacitance	F
C_{gs}, C_{gd}, C_{gb}	Gate to source, drain, body capacitances	F
C_{gso}, C_{gdo}, C_{gbo}	Overlap capacitances	F

5. Gate Capacitances (nMOS)

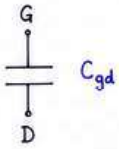
(a) C_{gs} (Gate to Source)

- Consists of oxide capacitance in series with depletion / inversion capacitance near source.
- Varies with inversion level.



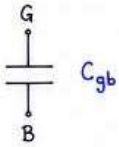
(b) C_{gd} (Gate to Drain)

- Consists of oxide capacitance in series with depletion / inversion capacitance near drain.



(c) C_{gb} (Gate to Body)

- Due to overlap of gate with source/drain diffusion and fringing fields.
- Usually modeled as C_{gbo} (overlap capacitance).



6. Capacitances in Different Regions (nMOS)

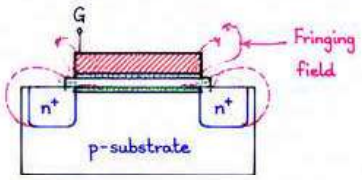
Region	Condition	C_{gs}	C_{gd}	C_{gb}
Cut-off	$V_{GS} \le V_T$	C_{gso} (overlap)	C_{gdo} (overlap)	C_{gbo}
Linear	$V_{GS} > V_T$, $V_{DS} < V_{GS} - V_T$	$\approx \frac{2}{3} C_{ox} W x + C_{gso}$	$\approx \frac{1}{3} C_{ox} W x + C_{gdo}$	C_{gbo}
Saturation	$V_{DS} \ge V_{GS} - V_T$	$\approx \frac{2}{3} C_{ox} W L_{eff} + C_{gso}$	$\approx 0 + C_{gdo}$	C_{gbo}

where x = channel charge centroid distance from source.

7. Oxide Capacitance with Fringing (Perimeter Capacitance)

- At the edges of gate, field lines fringe into silicon.
- Modeled as C_{oxp} (F/cm) along the gate perimeter.
- Total overlap capacitance :

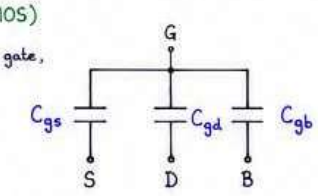
$$C_{gbo} = C_{oxp}(L + 2W)$$



8 Total Capacitance Summary (nMOS)

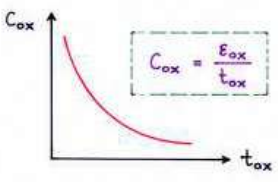
Total gate capacitance seen looking into gate,

$$C_G = C_{gs} + C_{gd} + C_{gb}$$



9 Effect of Oxide Thickness

- $C_{ox} \propto \frac{1}{t_{ox}}$
- Thinner oxide ⇒ Higher C_{ox}
- Improves drive current but increases gate leakage.



☑ Key Takeaways

- ☑ Capacitances determine speed, delay and AC behaviour.
- ☑ Oxide capacitance C_{ox} is fundamental and inversely proportional to oxide thickness.
- ☑ Junction capacitances are bias dependent.
- ☑ Gate capacitances vary with region (cut-off, linear, saturation).
- ☑ Proper modeling is essential for high performance design.

Author : Konda Tharun Vivekananda

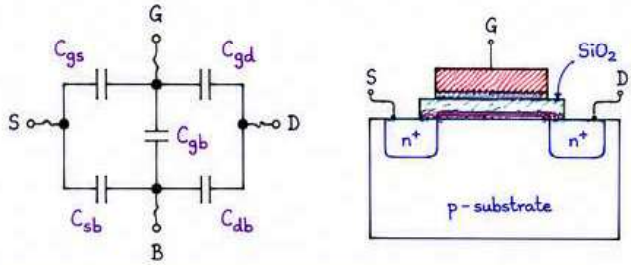
CMOS – An Industry Perspective

1. MOSFET Capacitance – Lumped Representation and its explanation of each capacitence

☑ Introduction :

- MOSFET capacitances model the charge stored in oxide and depletion/inversion regions.
- For small signal analysis, they are represented by a lumped capacitance network.

A. Lumped Capacitance Representation (nMOS)



B. Explanation of Each Capacitance

1. C_{gs} (Gate to Source)
   - Consists of oxide capacitance between gate and source overlap + fringing, and depletion/inversion capacitance in channel region near source.
   - High in accumulation, reduces in depletion.
2. C_{gd} (Gate to Drain)
   - Consists of oxide capacitance between gate and drain overlap + fringing, and depletion/inversion capacitance in channel near drain.
   - Strongly voltage dependent.
3. C_{gb} (Gate to Body)
   - Due to overlap of gate with body (fringing field).
   - Nearly constant, small value.
4. C_{sb} (Source to Body)
   - Depletion capacitance of p-n junction (source to body).
   - Decreases as V_{SB} increases (reverse bias).
5. C_{db} (Drain to Body)
   - Depletion capacitance of p-n junction (drain to body).
   - Decreases as V_{DB} increases.

2. Approximate Oxide Capacitance Value Formulaes

- Gate oxide capacitance per unit area (C_{ox}) :

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \quad (F/cm^2)$$

where, $\varepsilon_{ox} = \varepsilon_0 \varepsilon_{r,ox}$

$\varepsilon_0 = 8.854 \times 10^{-14}$ F/cm, $\varepsilon_{r,ox} \approx 3.9$ (SiO_2)

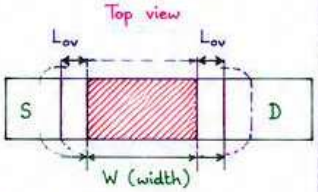
- Oxide capacitance for an area A :

$$C_{ox} = \frac{\varepsilon_{ox} A}{t_{ox}} \quad (F)$$

- Overlap capacitance :

$$C_{ov} = C_{ox} \cdot W \cdot L_{ov}$$

Top view



- Fringing (perimeter) capacitance :

$$C_{fr} = C_{oxp} \cdot P$$

where P = Perimeter of gate

$C_{oxp} \approx (0.2 - 0.4)\, C_{ox}$ (F/cm)

- Total oxide capacitance between gate and a region R :

$$C_{gR,ox} \approx C_{ox} A_{ov} + C_{fr} P_{edge}$$

A_{ov} = Overlap area with R

P_{edge} = Edge perimeter facing R

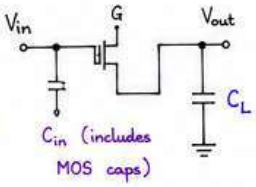
☑ Key Points :

- Oxide capacitances is inversely proportional to oxide thickness.
- Thinner oxide ⇒ Higher C_{ox} ⇒ Higher gate capacitances.
- Overlap + fringing contribute significantly in scaled devices.

3. Effects of MOSFET Capacitences

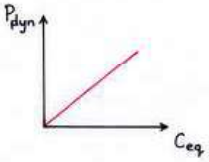
1. Speed & Delay

- Higher capacitances increase RC delay.
- Gate capacitance increases input load.
- C_{gd} (Miller effect) increases effective input capacitance.



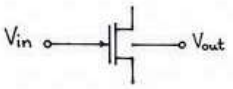
2. Switching Power

- Dynamic power : $P_{dyn} = \alpha\, C_{eq}\, V_{DD}^2\, f$
- Larger capacitances ⇒ Higher power.
- Reducing C_{ox} and overlaps reduces power.



3. Miller Effect (C_{gd})

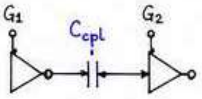
- During switching, C_{gd} is multiplied.
- $C_{in,eff} = C_{gs} + (1 + A_v) C_{gd}$
- Increases delay and reduces bandwidth.



$A_v = -g_m (R_D \| R_L)$

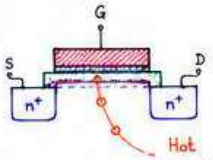
4. Charge Sharing & Crosstalk

- Capacitances cause charge sharing during transitions.
- Can cause glitches in high speed circuits.



5. Reliability

- Large gate capacitance ⇒ more charge in oxide.
- Can affect hot carrier stress and oxide reliability.



4. Capacitances in Different Regions (nMOS)

Region	Condition	C_{gs}	C_{gd}	C_{gb}	C_{sb}	C_{db}
Cut-off	$V_{GS} \le V_T$	C_{gso} (ov)	C_{gdo} (ov)	C_{gbo}	C_{jo}	C_{jo}
Linear	$V_{GS} > V_T$, $V_{DS} < V_{GS} - V_T$	$\approx \frac{2}{3} C_{ox} W x + C_{gso}$	$\approx \frac{1}{3} C_{ox} W x + C_{gdo}$	C_{gbo}	C_j (rev)	C_j (rev)
Saturation	$V_{DS} \ge V_{GS} - V_T$	$\approx \frac{2}{3} C_{ox} W L_{eff} + C_{gso}$	$\approx C_{gdo}$	C_{gbo}	C_j (rev)	C_j (rev)

x = channel charge centroid distance from source.

C_{jo} = zero bias junction cap.

C_j (rev) = reverse biased junction cap.

5. Summary of Important Capacitance Expressions

Capacitance	Expression
Gate Oxide Capacitance per unit area	$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$
Oxide Capacitance for area A	$C_{ox} = \frac{\varepsilon_{ox} A}{t_{ox}}$
Overlap Capacitance	$C_{ov} = C_{ox} W L_{ov}$
Fringing Capacitance	$C_{fr} = C_{oxp} P$
Gate to Source (general)	$C_{gs} = C_{gs,ox} + C_{dep/inv}$
Gate to Drain (general)	$C_{gd} = C_{gd,ox} + C_{dep/inv}$
Gate to Body	$C_{gb} = C_{gbo}$ (overlap/fringing)
Source to Body	$C_{sb} = C_j$ (p-n junction)
Drain to Body	$C_{db} = C_j$ (p-n junction)

☑ Overall Takeaway

- MOS capacitances arise from oxide, depletion and inversion.
- They are voltage dependent and strongly affect speed, power and reliability.
- Proper modeling and minimization are essential in high performance CMOS design.

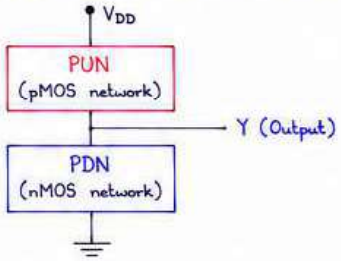
Author : Konda Tharun Vivekananda

CMOS - An Industry Perspective

1. CMOS Basic Gates Design - Intro, why to design, PUN, PDN, Static Design and Dynamic Design

☑ Introduction :

- CMOS gates are designed using complementary networks.
- nMOS forms Pull-Down Network (PDN) connected to GND.
- pMOS forms Pull-Up Network (PUN) connected to V_{DD}.
- Output node is the common node between PUN and PDN.

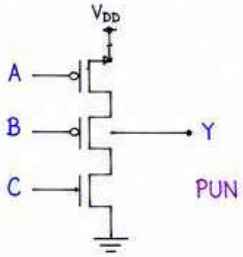


2. Why to Design CMOS Gates ?

- High noise immunity due to full rail-to-rail swing.
- Very low static power (ideally zero in steady state).
- High fan-out capability.
- Scalability and high packing density.
- Low power and high speed operation.

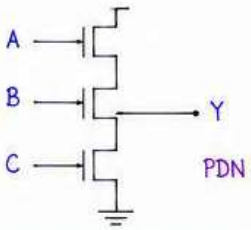
3. Pull-Up Network (PUN)

- Built using pMOS transistors.
- Connected between V_{DD} and output.
- Conducts when output should be logic '1'.
- Implements the dual (complement) of the logic function.
- pMOS turns ON when input = '0'.



4. Pull-Down Network (PDN)

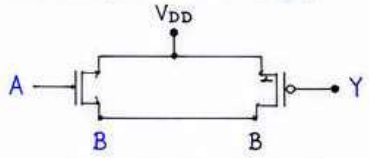
- Built using nMOS transistors.
- Connected between output and GND.
- Conducts when output should be logic '0'.
- Implements the true logic function.
- pMOS turns ON when input = '1'.



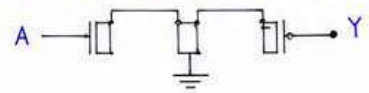
Example : 2-Input NAND Gate

Logic Function : $Y = \overline{A \cdot B}$

PUN (complement = $A' + B'$)



PDN (true function = $A \cdot B$)



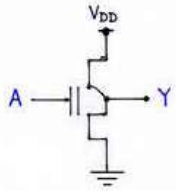
Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

5. Static Design

- Both PUN and PDN are present.
- At any steady input condition, either PUN or PDN conducts.
- No direct path from V_{DD} to GND.
- Ideally zero static power.

Inverter (Static)

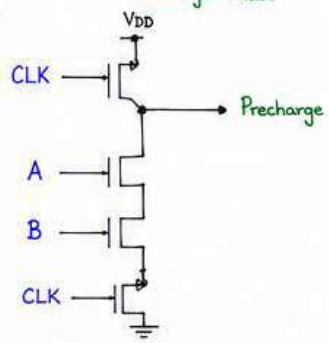


A	Y
0	1
1	0

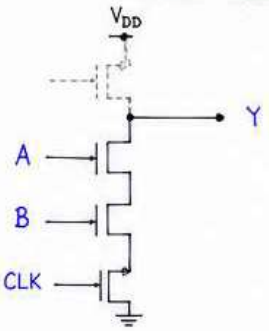
6. Dynamic Design

- Only one network is pre-charged; the other evaluates.
- Requires clock (CLK).
- Lower transistor count → lower area and capacitance.
- Higher speed but leakage and noise sensitive.
- Types :
  - Domino Logic
  - Pass Transistor Logic
  - Transmission Gate Logic

Precharge Phase



Evaluate Phase



7. PUN vs PDN Relationship

Feature	PUN (pMOS)	PDN (nMOS)
Connected between	V_{DD} and output	Output and GND
Conducts when	Output = 1	Output = 0
Transistor ON when	Input = 0	Input = 1
Implements	Dual / Complement	True logic function
Design rule	Parallel for OR Series for AND	Series for AND Parallel for OR

8. Static vs Dynamic Design - Comparison

Feature	Static Design	Dynamic Design
Networks	PUN + PDN always present	One network pre-charged, other evaluates
Clock	Not required	Required
Power (static)	Ideally zero	Some leakage in precharge device
Noise immunity	High	Lower
Speed	Moderate	High
Transistor count	Higher	Lower
Use cases	General purpose, robust	High performance, low capacitance designs

9. Key Takeaways

- ☑ CMOS logic uses complementary PUN and PDN.
- ☑ PUN implements complement of logic function; PDN implements true function.
- ☑ Static design is robust with zero static power.
- ☑ Dynamic design is faster and smaller but needs clock and is more sensitive to leakage and noise.
- ☑ Proper choice of design style depends on speed, power, area and robustness requirements.

Author : Konda Tharun Vivekananda

CMOS - An Industry Perspective

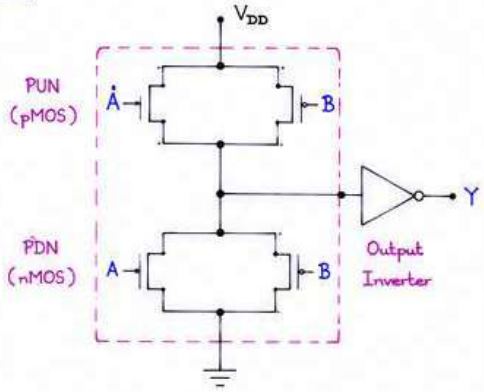
1. CMOS Basic Gates Design - AND design, Steps to design, properties, Working, problems in CMOS Design

1. AND Gate Design

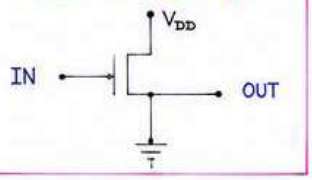
- Logic : $Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

CMOS Implementation (AND gate)
(PUN + PDN followed by Inverter)



Inverter (used at output)



Function : $OUT = \overline{IN}$

2. Steps to Design (AND Gate)

1. Write the logic function. ($Y = A \cdot B$).
2. Determine when output = 0 → design PDN. (PDN ON when output = 0)
3. Determine when output = 1 → design PUN. (PUN ON when output = 1)
4. Ensure duality (series ↔ parallel, nMOS ↔ pMOS).
5. Connect PDN to GND and PUN to V_{DD}.
6. Feed the output node to a CMOS inverter.
7. Join inverter output as final Y.

3. Properties

- Full rail-to-rail output (0 to V_{DD}).
- Ideally zero static power.
- High noise immunity.
- Symmetric rise and fall paths (if sized properly).

4. Working (AND Gate)

A	B	PDN (nMOS)	PUN (pMOS)	After Inv. (Output Y)	Meaning
0	0	OFF (open)	ON (series)	0	0.AND.0 = 0
0	1	OFF (open)	ON (series)	0	0.AND.1 = 0
1	0	OFF (open)	ON (series)	0	1.AND.0 = 0
1	1	ON (parallel)	OFF (series)	1	1.AND.1 = 1

5. Problems in CMOS Design

- Area : PUN has pMOS (wider) → larger area.
- Power : Switching power ($P = \alpha C_L V_{DD}^2 f$).
- Delay : Series nMOS in PDN / PUN increases R_{eq} → delay ↑.
- Logical effort due to unequal rise / fall.
- Parasitic capacitances increase with more transistors.
- Short channel effects in scaled technologies.
- Layout issues : diffusion sharing, well taps, matching.

✳ Key Point :
AND gate is realized as NAND network (PUN series, PDN parallel) followed by inverter.
Inverter at output is essential in CMOS logic.

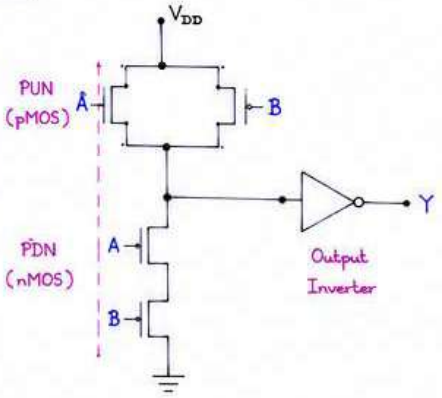
2. CMOS Basic Gates Design - OR design, Steps to design, properties, Working, problems in CMOS Design

1. OR Gate Design

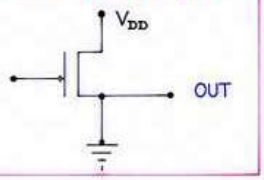
- Logic : $Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

CMOS Implementation (OR gate)
(PUN + PDN followed by Inverter)



Inverter (used at output)



Function : $OUT = \overline{IN}$

2. Steps to Design (OR Gate)

1. Write the logic function. ($Y = A + B$).
2. Determine when output = 0 → design PDN. (PDN ON when output = 0)
3. Determine when output = 1 → design PUN. (PUN ON when output = 1)
4. Ensure duality (series ↔ parallel, nMOS ↔ pMOS).
5. Connect PDN to GND and PUN to V_{DD}.
6. Feed the output node to a CMOS inverter.
7. Join inverter output as final Y.

3. Properties

- Full rail-to-rail output.
- Ideally zero static power.
- High noise margin.
- Lower logical effort for OR in PDN (parallel nMOS).

4. Working (OR Gate)

A	B	PDN (nMOS)	PUN (pMOS)	After Inv. (Output Y)	Meaning
0	0	OFF (series)	ON (parallel)	0	0+0 = 0
0	1	ON (series)	OFF (parallel)	1	0+1 = 1
1	0	ON (series)	OFF (parallel)	1	1+0 = 1
1	1	ON (series)	OFF (parallel)	1	1+1 = 1

5. Problems in CMOS Design

- Area : More transistors → larger layout.
- Power : Dynamic power increases with load & frequency.
- Delay : Series nMOS in PDN increases R_{eq} → delay ↑.
- Rise / fall asymmetry (pMOS weaker than nMOS).
- Parasitic capacitance affects speed and power.
- Leakage power in advanced nodes.
- Layout matching, well proximity & latch-up concerns.

✳ Key Point :
OR gate is realized as NOR network (PUN parallel, PDN series) followed by inverter.
Either AND or OR alone can implement any logic.
Output inverter gives full swing and low static power.

Author : Konda Tharun Vivekananda

CMOS - An Industry Perspective

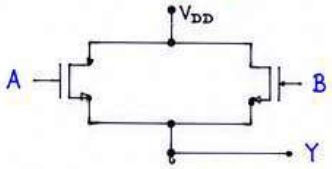
1. CMOS Basic Gates Design - NAND design, Steps to design, properties, CMOS NAND Working, problems in CMOS Design

1. NAND Gate Design

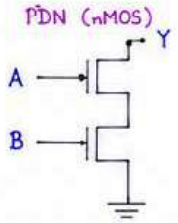
- Logic : $Y = \overline{A \cdot B}$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

PUN (pMOS)



PDN (nMOS)



2. Steps to Design (NAND Gate)

1. Write the logic function.
2. Determine when output = 0 → design PDN. (PDN ON when output = 0)
3. Determine when output = 1 → design PUN. (PUN ON when output = 1)
4. Ensure duality (series ↔ parallel, nMOS ↔ pMOS).
5. Connect PDN to GND and PUN to V_{DD}.
6. Join output nodes of PUN and PDN.

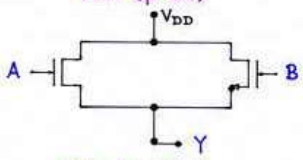
3. Properties

- Full rail-to-rail output (0 to V_{DD}).
- Ideally zero static power.
- High noise immunity.
- Symmetric rise and fall paths (if sized properly).

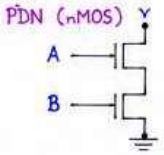
4. CMOS NAND Gate - Working (example)

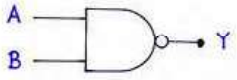
A	B	PDN (nMOS)	PUN (pMOS)	Output Y
0	0	OFF (open)	ON (parallel)	1 (V_{DD})
0	1	OFF (open)	ON (parallel)	1
1	0	OFF (open)	ON (parallel)	1
1	1	ON (series)	OFF (open)	0 (GND)

PUN (pMOS)



PDN (nMOS)





Symbol : NAND

5. Problems in CMOS Design

- Area : PUN has pMOS (wider) → larger area.
- Power : Switching power ($P = \propto C_L V_{DD}^2 f$).
- Delay : Series nMOS in PDN increases R_{eq} → delay ↑.
- Logical effort due to unequal rise/fall.
- Parasitic capacitances increase with more transistors.
- Short channel effects in scaled technologies.
- Layout issues : diffusion sharing, well taps, matching.

> ✳ Key Point :
> NAND gate is a universal gate and forms the basis for most digital designs.

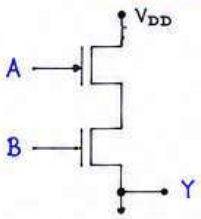
2. CMOS Basic Gates Design - NOR design, Steps to design, properties, CMOS NAND Working, problems in CMOS Design

1. NOR Gate Design

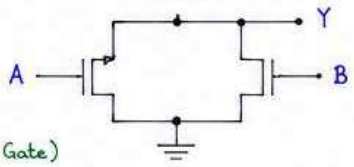
- Logic : $Y = \overline{A + B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

PUN (pMOS)



PDN (nMOS)



2. Steps to Design (NOR Gate)

1. Write the logic function.
2. Determine when output = 0 → design PDN. (PDN ON when output = 0)
3. Determine when output = 1 → design PUN. (PUN ON when output = 1)
4. Ensure duality (series ↔ parallel, nMOS ↔ pMOS).
5. Connect PDN to GND and PUN to V_{DD}.
6. Join output nodes of PUN and PDN.

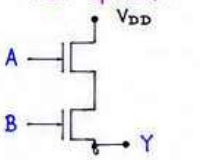
3. Properties

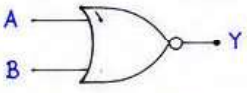
- Full rail-to-rail output.
- Ideally zero static power.
- High noise margin.
- Lower logical effort for OR in PDN (parallel nMOS).

4. CMOS NOR Gate - Working (example)

A	B	PDN (nMOS)	PUN (pMOS)	Output Y
0	0	OFF (open)	ON (series)	1 (V_{DD})
0	1	ON (one path)	OFF (break)	0 (GND)
1	0	ON (one path)	OFF (break)	0
1	1	ON (parallel)	OFF (break)	0

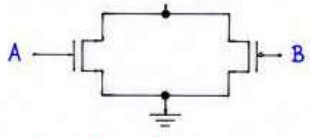
PUN (pMOS)





Symbol : NOR

PDN (nMOS)



5. Problems in CMOS Design

- Area : More transistors → larger layout.
- Power : Dynamic power increases with load & frequency.
- Delay : Series pMOS in PUN increases R_{eq} → slow rise.
- Rise and fall delay asymmetry (pMOS weaker than nMOS).
- Parasitic capacitances affect speed and power.
- Leakage power in advanced nodes.
- Layout matching, well proximity & latch-up concerns.

> ✳ Key Point :
> NOR gate is a universal gate and is dual of NAND.
> Either NAND or NOR alone can implement any logic.

Author : Konda Tharun Vivekananda

CMOS - An Industry Perspective

1. CMOS - Problems in CMOS and Alternatives such as Ratioed Logic

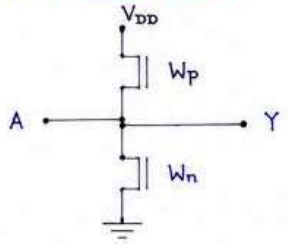
1. Problems in CMOS

- High transistor count → Large area. (Both PUN and PDN required.)
- Higher input capacitance → Higher dynamic power.
- Series devices in PUN/PDN → Increased resistance, slower switching.
- Short-circuit (crowbar) current during switching → extra power.
- Leakage in advanced nodes → standby power ↑.
- Layout complexity → more routing, matching issues.

2. Alternatives : Ratioed Logic

- Uses only nMOS network + load (pMOS or resistor).
- Fewer transistors → smaller area, lower input cap.
- Ratio of pull-up/pull-down controls logic levels.
- Reduced static power but suffers degraded levels.
- Not full swing; noise margin is limited.

Example : Ratioed Inverter



Operation

A	nMOS	Y
0	OFF	1 (weak)
1	ON	0 (strong)

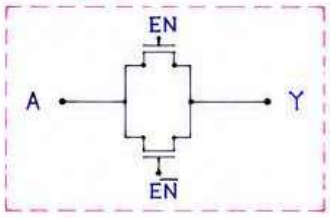
Note : Logic levels depend on Wp / Wn ratio.

3. CMOS - Problems in CMOS and Alternatives such as Transmission Gate

1. Transmission Gate (TG)

- Parallel combination of nMOS and pMOS controlled by complementary signals.
- Passes both '0' and '1' strongly (full swing).
- Low on-resistance → better speed.
- Zero static power (only leakage).
- Used in multiplexers, flip-flops, level shifters.

Circuit Symbol



Truth Table

EN	$\overline{EN}$	A	Y
0	1	X	Z (Hi-Z)
1	0	0	0
1	0	1	1

Operation :

- EN = 1 → nMOS ON, pMOS OFF → node Y connected to A.
- EN = 0 → pMOS ON, nMOS OFF → node Y connected to A.
- Provides rail-to-rail output (no V_{TH} loss).

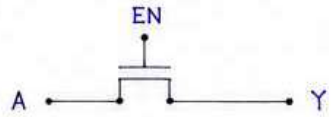
* Key Point : TG solves the problem of weak logic levels in pass Transistor logic by using both nMOS and pMOS.

2. CMOS - Problems in CMOS and Alternatives such as Pass Transistor Logic

1. Pass Transistor Logic (PTL)

- Uses single nMOS (or pMOS) to pass logic.
- Reduces transistor count and input capacitance.
- No static power (only leakage).
- Limitation : Weak '0' or weak '1' due to V_{TH} drop. (nMOS passes '0' weak, pMOS passes '1' weak)
- Not suitable for cascading many stages.

Circuit Example : nMOS Pass Gate (Buffer)



Operation (nMOS PTL)

EN	A	Y
0	X	Z (Hi-Z)
1	0	0 (weak)
1	1	1 (strong)

Note :

Weak '0' level ≈ 0V , Weak '1' level ≈ $V_{DD} - V_{THn}$

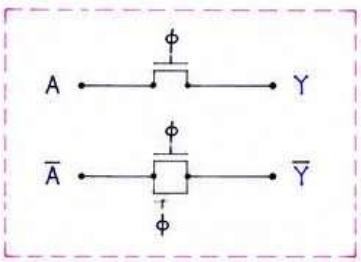
Applications : Multiplexers, level shifters, latches (with care)

4. CMOS - Problems in CMOS and Alternatives such as Differential Pass Transistor Logic

1. Differential Pass Transistor Logic (DPTL)

- Uses complementary (differential) signals and pass devices.
- Always maintains full-swing logic (no threshold loss).
- Lower power and transistor count compared to static CMOS.
- Less sensitive to noise than single-ended PTL.
- Used in high-speed multiplexers, latches, sense amplifiers.

Basic Structure



ϕ = Clock / Enable (Same for both paths)

Outputs are complementary : $\overline{Y} = \overline{Y}$

Operation

- When ϕ = 1 → inputs passed to outputs.
- When ϕ = 0 → outputs hold previous values (isolated).
- Differential nature ensures full-swing levels.

* Key Point : DPTL provides high-speed operation with low power and full logic levels by using complementary pass networks.

Author : Konda Tharun Vivekananda

CMOS - An Industry Perspective

1. CMOS - Logic Effort

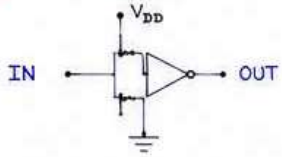
1.1 Definition

- Logic effort is a simple metric used to estimate the delay of a logic gate driving a load.
- It separates the effect of electrical effort from the intrinsic difficulty of the gate.
- It allows quick comparison of different logic styles independent of technology.

1.2 Basic Terms

- g (logic effort) : Delay relative to a reference inverter when driving the same load.
- h (electrical effort) : Ratio of actual load to the input capacitance of the driving gate.
  $h = C_L / C_{in}$
- p (parasitic delay) : Delay due to internal diffusion and wiring, independent of load.
- d (stage delay) : $d = g \times h + p$
- Path delay : : $D \; \Sigma \; (g_i h_i + p_i)$

1.3 Reference Inverter



For reference inverter
$g_{inv} = 1$
$p_{inv} = 1$
$C_{in,inv} = C_{g,n} + C_{g,p} = 2C$

2. Calculation with Examples

2.1 Logic Effort Values for Common Gates

Gate	Symbol	g (logic effort)	p (parasitic delay)
Inverter		1	1
2-input NAND		4/3	2
2-input NOR		5/3	2
2-input AND		4/3	2
2-input OR		5/3	2
2-input XOR		4	4

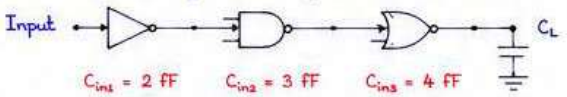
2.2 Example - 1

A 2-input NAND gate drives a load $C_L = 10$ fF.
Gate input capacitance $C_{in} = 3$ fF.
Given : For NAND → $g = 4/3$, $p = 2$
1) Electrical effort : $h = C_L / C_{in} = 10/3 \approx 3.33$
2) Stage delay : $d = g \times h + p = (4/3) \times 3.33 + 2$
$\approx 4.44 + 2 = 6.44$ (in units of τ)

* Note :
- Actual delay = $d \times \tau$
- τ = technology dependent time constant.

2.3 Example - 2

A chain of Inverter → NAND → NOR drives $C_L = 20$ fF.
Given : $C_{in(inv)} = 2$ fF , $C_{in(NAND)} = 3$ fF , $C_{in(NOR)} = 4$ fF.
$g_{inv} = 1$, $g_{NAND} = \frac{4}{3}$, $g_{NOR} = \frac{5}{3}$, $p_{inv} = 1$, $p_{NAND} = 2$, $p_{NOR} = 2$



Electrical efforts : $h_1 = \frac{3}{2} = 1.5$, $h_2 = \frac{4}{3} \approx 1.33$, $h_3 = \frac{20}{4} = 5$
Stage delays : $d_1 = 1 \times 1.5 + 1 = 2.5$
$d_2 = (4/3) \times 1.33 + 2 \approx 3.78$
$d_3 = (5/3) \times 5 + 2 \approx 10.33$
Path delay : $D = d_1 + d_2 + d_3 \approx 16.61$ (in units of τ)

3. Advantages and Applications

3.1 Advantages

- Provides technology-independent delay estimation.
- Helps in comparing different logic styles.
- Guides optimal effort distribution in a logic path.
- Useful for preliminary sizing and architecture evaluation.
- Simple hand calculations with good accuracy.

3.2 Applications

- Estimating delay in combinational logic paths.
- Gate sizing and effort optimization.
- Selecting logic styles in high-speed design.
- Early stage performance analysis in ASIC / SoC design.
- Useful in synthesis, place & route and timing closure.

General Delay Model

$$d \text{ (stage delay)} = g \times h + p$$

$$D \text{ (path delay)} = \sum_{i=1}^{N} (g_i h_i + p_i)$$

$$h = C_{out(prev)} / C_{in(next)}, \quad f = \prod^{n} h_i$$

Where,
- g → logic effort
- h → electrical effort
- p → parasitic delay
- f → path electrical effort

4. Logic Effort Summary

4.1 Key Points

- Delay ∝ (gate effort) × (electrical effort) + parasitic.
- Logic effort separates intrinsic gate property (g) from load effect (h).
- Reference inverter has g = 1, p = 1.
- Total path delay is the sum of individual stage delays.

4.2 Comparison of Gates

Gate	Pull-up Network	Pull-down Network	g	p	Comments
Inverter	1	1	1	1	Reference gate
NAND2	Parallel	Series	4/3	2	Fast
NOR2	Series	Parallel	5/3	2	Slightly slower than NAND2
AND2	Series	Parallel	4/3	2	Same g as NAND2
OR2	Parallel	Series	5/3	2	Same g as NOR2
XOR2	Complex	Complex	4	4	High effort gate

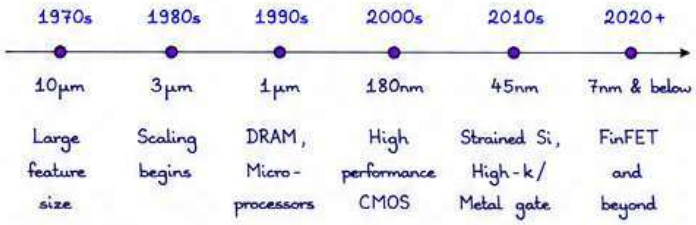
Quick Flow for Logic Effort Analysis
1. Identify gate type for each stage → get g and p.
2. Determine C_{in} of each gate.
3. Compute electrical effort $h = C_{out(prev)} / C_{in(next)}$.
4. Calculate stage delay $d = g \times h + p$.
5. Add all stage delays → total path delay D.

Author : Konda Tharun Vivekananda

CMOS - An Industry Perspective

1. CMOS - Evolution, Why MOSFET become difficult.

1.1 Evolution of CMOS Technology



1.2 Why MOSFET become difficult at advanced nodes

- Short channel effects : DIBL, threshold roll-off.
- Poor electrostatic control : Gate controls channel only from one side.
- Gate leakage current increases drastically.
- Higher OFF state leakage (I_{off}) → more power.
- Variability issues : V_{th} variation, line edge roughness.
- Drive current (I_{on}) saturation → scaling no longer effective.
- Thermal issues due to high power density.
- More complex process : Multi-patterning, EUV, etc.
- Thus, planar MOSFET scaling becomes challenging.

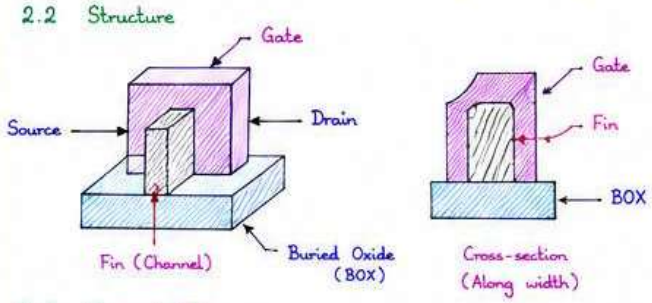
> Key Takeaway :
> To overcome these issues, new device structures like FinFET are introduced for better gate control and improved performance at nanoscale.

2. FinFET Introduction

2.1 Birth

- Introduced around early 2000s for sub-45nm nodes.
- Commercially adopted from ~ 22nm technology node.
- Developed to overcome scaling limits of planar MOSFETs.
- Provides better electrostatic control by surrounding channel.

2.2 Structure



2.3 Characteristics

- Multi-gate device (usually 3-sided gate).
- Excellent gate control over channel.
- Lower SCE (Short Channel Effects).
- Higher drive current (I_{on}) for same power.
- Reduced leakage current (I_{off}).
- Better performance and lower power.

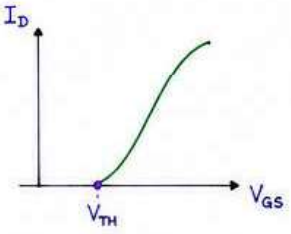
2.4 Working

- Voltage applied to Gate controls the charge in the Fin (channel).
- When $V_{GS} > V_{TH}$ → inversion layer forms on fin surfaces.
- Carriers flow from Source to Drain through the fin.
- Current conduction happens along the length of the fin.

2.5 Modes

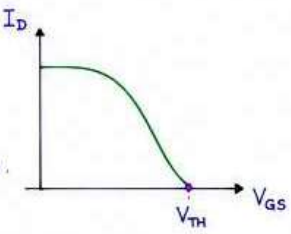
(a) Enhancement Mode (E-mode)

- Normally OFF at $V_{GS} = 0$.
- Conduction starts when $V_{GS} > V_{TH}$.
- Used in standard CMOS.



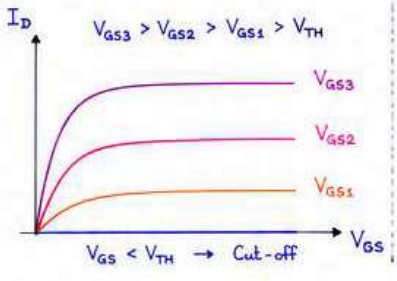
(b) Depletion Mode (D-mode)

- Normally ON at $V_{GS} = 0$.
- Conducts at zero gate bias.
- Turns OFF when $V_{GS} < V_{TH}$ (negative).
- Used in special low-power circuits.

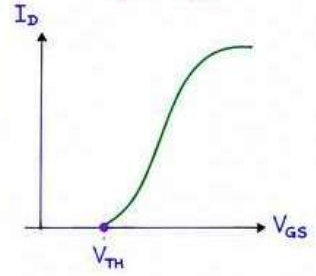


2.6 I-V Characteristics

(a) Output Characteristics (I_D vs V_{DS})



(b) Transfer Characteristics (I_D vs V_{GS})



2.7 Benefits

- Better electrostatic control.
- Higher drive current (I_{on}).
- Lower leakage current.
- Reduced short channel effects.
- Higher switching speed.
- Lower dynamic power.

2.8 Limitations

- More complex fabrication.
- Higher process cost.
- Variability in fin width/height.
- Parasitic resistance increase.
- Heat dissipation challenges at advanced nodes.

2.9 Applications

- Advanced CMOS technologies (22nm, 16nm, 10nm, 7nm, 5nm).
- High performance microprocessors and SoCs.
- Low power mobile and IoT devices.
- High speed digital circuits.
- Analog/RF circuits requiring low leakage and high speed.

> Summary :
> FinFET is a multi-gate (typically 3-sided) device that provides strong gate control, reduces leakage and short channel effects, and enables continued CMOS scaling in advanced technology nodes.

CMOS - An Industry Perspective

1. GAA FET

1.1 Introduction

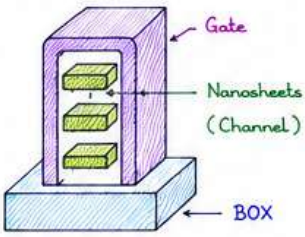
- Gate-All-Around FET (GAA FET) is the most advanced 3D MOSFET structure providing maximum electrostatic control.

1.2 Birth

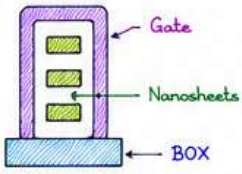
- Concept proposed in early 2010s.
- Commercially adopted in ~3nm node and beyond (from mid 2020s).

1.3 Structure

(a) 3D View



(b) Cross-section (Along width)



1.4 Characteristics

- Gate completely surrounds the channel from all sides (4-sided).
- Excellent electrostatic control & highest drive current.
- Very low SCE (Short Channel Effects).
- Lowest leakage current (I_{off}).
- Better subthreshold slope.
- Supports very small channel lengths (< 5nm).
- Scales well with nanosheet width and count.

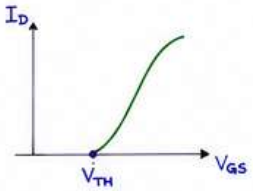
1.5 Working

- Voltage applied to gate controls the charge in the channel nanosheets.
- When $V_{GS} > V_{TH}$ → inversion charge forms on all surfaces of nanosheets.
- Carriers flow from Source to Drain through nanosheets.
- Conduction occurs along the length of the nanosheets.

1.6 Modes

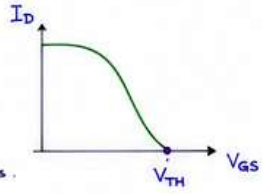
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- Used in standard CMOS.



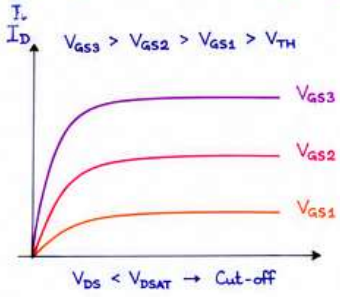
(b) Depletion Mode (D-mode)

- Normally ON at $V_{GS} = 0$.
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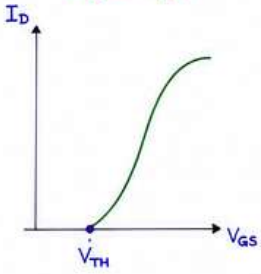


1.7 I-V Characteristics

(a) Output Characteristics (I_D vs V_{DS})



(b) Transfer Characteristics (I_D vs V_{GS})



2. Comparison of all - MOSFET, FINFET, GAA FET

Feature	Planar MOSFET	FinFET	GAA FET
Gate Control	1 side	3 sides	4 sides (complete)
Structure	Planar channel	Fin (vertical channel)	Nanosheets stack
Electrostatic Control	Weak	Good	Excellent
SCE	High	Low	Very Low
Leakage Current (I_{off})	High	Low	Very Low
Drive Current (I_{on})	Low	High	Highest
Subthreshold Slope (SS)	Poor	Better	Best
Scalability	Limited (<45nm)	Good (7nm-3nm)	Excellent (<3nm)
Power Efficiency	Low	Better	Best
Complexity	Low	Medium	High
Cost	Low	Medium	High
Nodes in Use	≥ 45nm	22nm - 3nm	3nm & below

3. Current Industry Technology, Current Semicon Trends and Technology

3.1 Current Industry Technology

- Leading foundries are in mass production of 3nm class nodes using GAA FET (nanosheet).
- High-NA EUV lithography, advanced materials and 3D stacking are widely used.
- GAA FET enables continued scaling beyond FinFET era.

3.2 Current Semicon Trends

- Continued scaling with GAA FET to sub-2nm nodes.
- More than Moore: 3D ICs, Chiplets, Heterogeneous integration.
- High-performance and low-power AI accelerators.
- Wide bandgap (SiC, GaN) for power & RF.
- Advanced packaging (CoWoS, InFO, Foveros).

3.3 Key Technology Directions

i) GAA nanosheet scaling
ii) New channel materials (SiGe, III-V, 2D materials)
iii) Novel interconnects (Low-k, Cu, Co)
iv) Power delivery & thermal management
v) AI/ML driven design and EDA automation

1.8 Benefits of GAA FET

- ✱ Ultimate electrostatic control.
- ✱ Highest drive current.
- ✱ Lowest leakage current.
- ✱ Excellent short channel immunity.
- ✱ Better subthreshold slope.
- ✱ Good scalability for future nodes.
- ✱ Lower dynamic power.
- ✱ Suitable for high density integrated circuits.

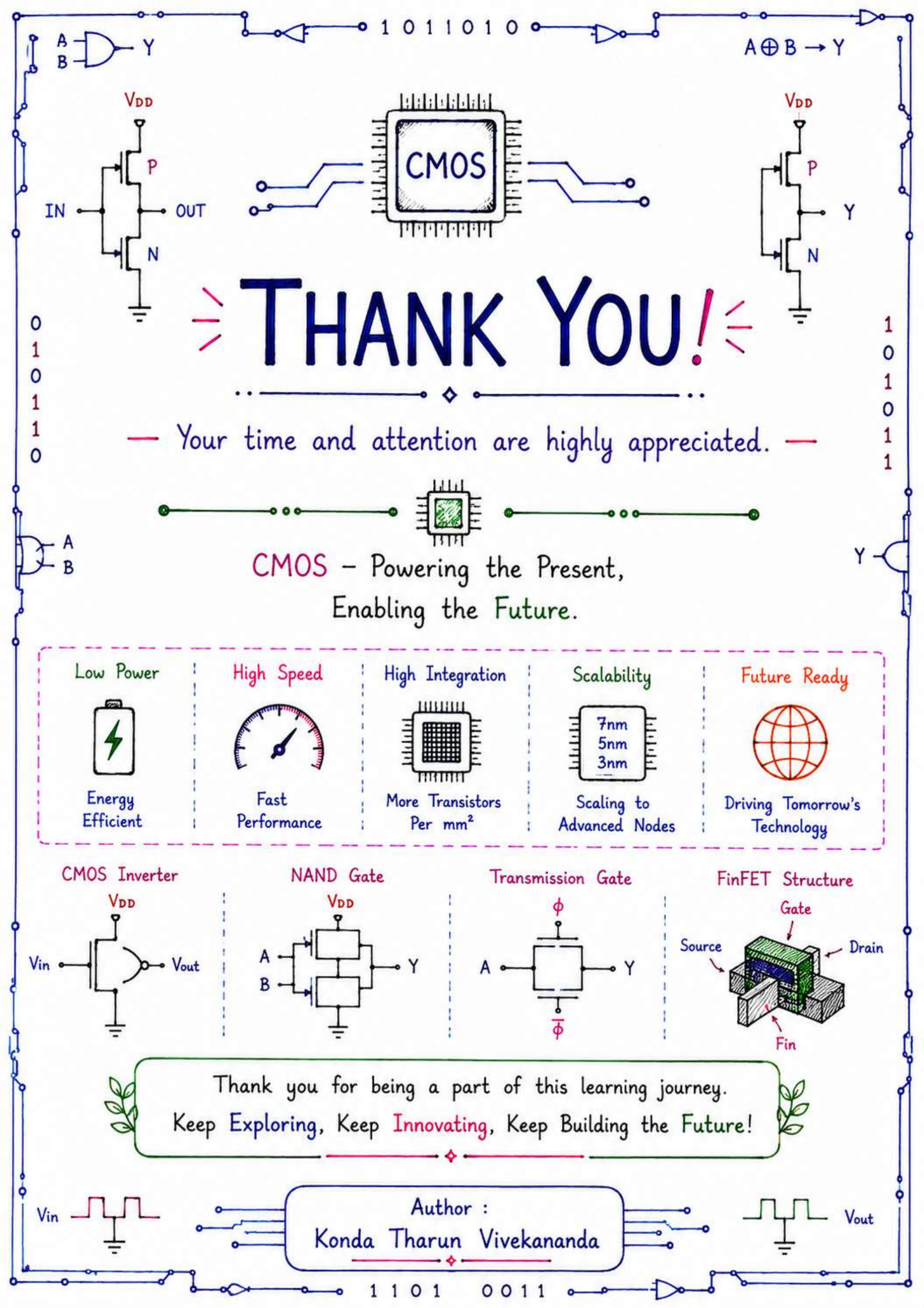
1.9 Limitations of GAA FET

- ✱ Complex 3D fabrication.
- ✱ Higher process cost.
- ✱ Precise nanosheet width and thickness control required.
- ✱ More sensitive to variability at nanoscale.
- ✱ Thermal management becomes challenging.

1.10 Applications

- ◆ High performance CPUs, GPUs and accelerators.
- ◆ Mobile SoCs and premium smartphones.
- ◆ AI/ML chips and data center processors.
- ◆ High density memory and logic.
- ◆ Future advanced digital and analog/RF circuits.

Author : Konda Tharun Vivekananda


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Thank you for being a part of this learning journey.
Keep Exploring, Keep Innovating, Keep Building the Future!
Vin
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